

CYBER ELIAS ACADEMY

TEACHING MANUAL

Data Entry

Session Roadmap · Lesson Notes · Concept Diagrams · Answer Keys

READY TO TEACH

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OVERVIEW

Course Roadmap

What we cover in every session

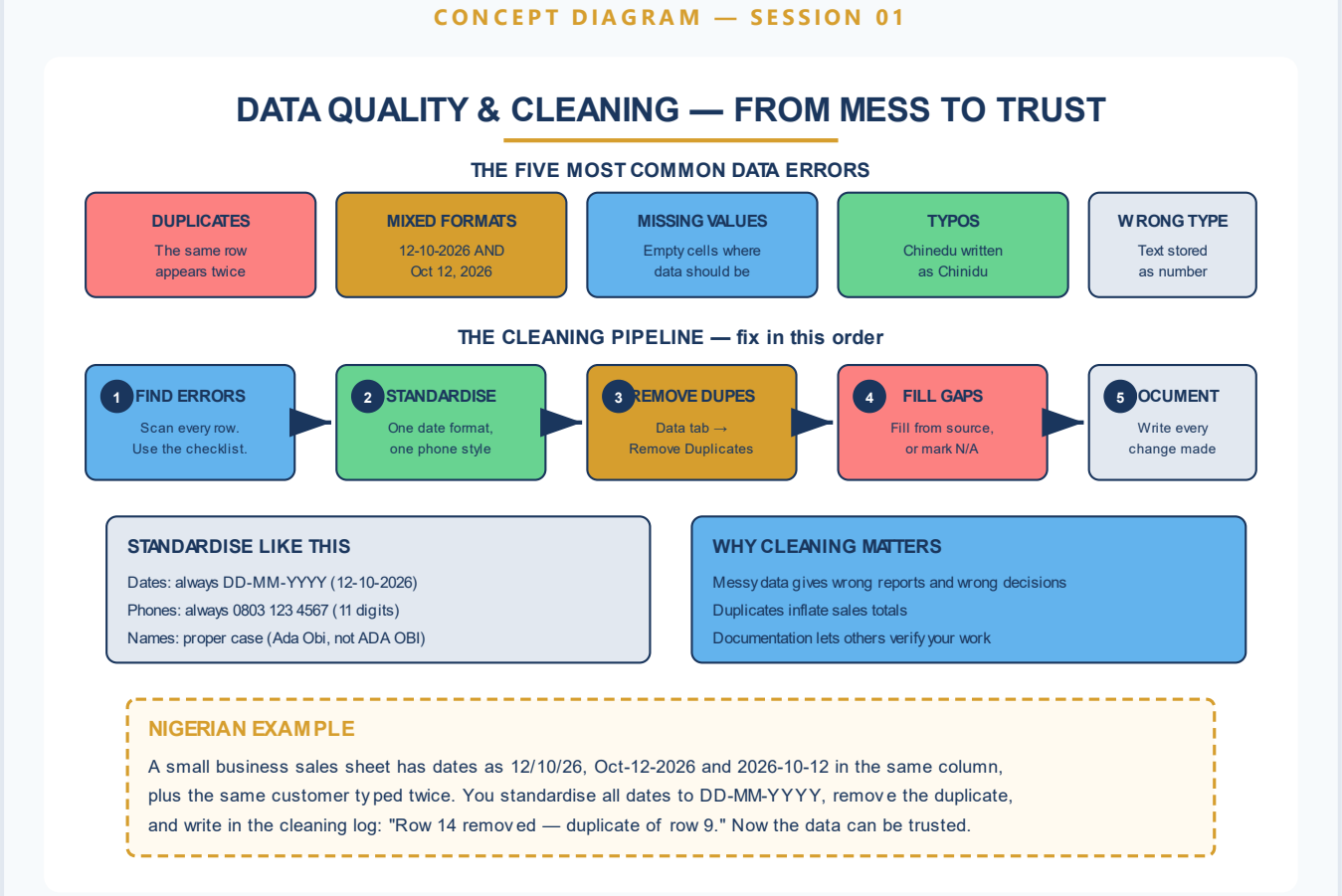
#	SESSION TOPIC	WEEK / MODULE	DURATION
01	Introduction to Data Entry		105 minutes
02	Spreadsheet Data Entry		105 minutes
03	Data Quality and Cleaning		105 minutes
04	Data Validation		105 minutes
05	Reconciliation Documentation		105 minutes
06	Final Project Capstone		105 minutes

How to use this book: Each session below is a complete, ready-to-teach lesson note. Section A tells you what to prepare, B lists the outcomes, D–F are the teaching content, G–H are your demos and practicals, and the knowledge check with answer key is at the end. The concept diagram for each session is embedded as a visual aid.

SESSION 01 OF 6

Introduction to Data Entry

CONCEPT DIAGRAM — SESSION 01



SESSION 1: INTRODUCTION TO DATA ENTRY

Week: 1 **Module:** Data Entry Fundamentals **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Basic computer skills (Typing & Computer Basics recommended) **Connection to Previous Session:** First session of Data Entry **Preparation Required Before Class:**

- Ensure Microsoft Excel is installed and functional on all machines
- Prepare typing test software or website (e.g., typingclub.com, 10fastfingers.com)
- Print "Data Types Reference" handout (one per learner)
- Prepare a sample dataset for demonstration

A. SESSION IDENTIFICATION

Field	Detail
Session Number	1
Title	Introduction to Data Entry

Field	Detail
Week	1
Module	Data Entry Fundamentals
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Basic computer skills
Connection to Previous Session	First session
Preparation	Typing test ready, handouts printed

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Define data entry** and explain its importance in business.
2. **Identify the four main data types:** text, numbers, dates and currency.
3. **Distinguish between data entry and data management.**
4. **Explain professional standards** for accuracy, consistency and confidentiality.
5. **Measure their typing speed** in words per minute (WPM) and accuracy percentage.
6. **Explain why** attention to detail matters in data entry.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for instructor demonstration

Software:

- Microsoft Excel 2019 or Microsoft 365
- Typing test website (e.g., 10fastfingers.com) or software

Printed Materials:

- "Data Types Reference" handout (one per learner)

Internet: Required for the typing test website.

Alternative for Learners Without Internet: Use a local typing test tool or a timed typing exercise in Notepad.

D. OPENING ACTIVITY (10 minutes)

"The Cost of a Mistake"

Instructor Setup: Write two versions of a phone number on the whiteboard: one correct and one with a single digit wrong.

Instructor Script: "Good morning, everyone. Welcome to the Data Entry course. Before we start, look at these two phone numbers. [Point to the whiteboard.] They look almost the same. But one digit is different. If this is a customer's phone number

and you enter it wrong, what happens? [Pause for answers.]

The customer does not receive a call. The business loses a sale. The customer is frustrated. All because of one wrong digit.

In data entry, accuracy is everything. A single wrong digit can cost a business money, time and reputation. This course will teach you to enter data with 100% accuracy — every time.

Let us begin."

Expected Learner Response: Recognition that small errors have big consequences. This sets the context for the session.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, let us make sure we are all on the same page. Quick review of basic computer skills."

Review Questions and Expected Answers:

1. "How do you open Microsoft Excel?"
   - Click the Start button, type "Excel," and click on Microsoft Excel.
2. "How do you create a new spreadsheet?"
   - Click "File," then "New," then "Blank workbook." Or press Ctrl+N.
3. "How do you save a file?"
   - Press Ctrl+S or click File > Save.
4. "What is a cell in Excel?"
   - The intersection of a row and a column. Each cell has a unique reference (e.g., A1, B2).
5. "How do you select multiple cells?"
   - Click and drag over the cells. Or click the first cell, hold Shift, and click the last cell.

How to Address Common Gaps:

- If a learner cannot open Excel, guide them through the Start menu.
- If a learner forgot how to save, remind them: Ctrl+S.

F. FULL LECTURE NOTES (35 minutes)

Topic 1: What Is Data Entry? (5 minutes)

Instructor Script: "Data entry is the process of inputting information into a computer system. It can be typing text, numbers, dates or any other data into a spreadsheet, database or other software.

Where Is Data Entry Used?

- Banks — entering customer transactions
- Hospitals — entering patient records
- Schools — entering student grades
- Offices — entering sales records, inventory, customer information
- Government — entering census data, tax records
- E-commerce — entering product details, orders

Why Does Data Entry Matter? Data is the foundation of every business decision. If the data is wrong, the decisions based on that data will be wrong. Accurate data entry ensures that businesses have reliable information to work with."

Topic 2: Types of Data (10 minutes)

Instructor Script: "In data entry, you will work with four main types of data. Each type has its own rules and formatting requirements.

1. Text Data: Any combination of letters, numbers and symbols. Examples: names, addresses, descriptions.

- Always check spelling carefully.
- Be consistent with capitalisation (e.g., always use Title Case for names).

2. Numerical Data: Numbers used for calculations. Examples: quantities, prices, ages.

- Do not include currency symbols in numerical data (use formatting instead).
- Use the correct number of decimal places.

3. Date Data: Dates in a consistent format. Examples: 01/10/2026, 15/11/2026.

- Always use the same date format throughout the dataset.
- In Nigeria, the standard format is DD/MM/YYYY (day/month/year).

4. Currency Data: Monetary values. Examples: ₦15,000, ₦2,500.50.

- Always use the correct currency symbol.
- Use commas as thousand separators.
- Use the correct number of decimal places (usually 2 for kobo)."

Topic 3: Data Entry vs Data Management (5 minutes)

Instructor Script: "Data entry and data management are related but different.

Data Entry:

- Inputting data into a system
- Focused on accuracy and speed
- Following established formats and rules

Data Management:

- Organising, storing and maintaining data
- Includes data entry but also includes cleaning, validation, analysis and reporting
- Focused on data quality and usability

In this course, you will learn both — starting with data entry and moving into data management."

Topic 4: Professional Standards (10 minutes)

Instructor Script: "Professional data entry requires:

1. Accuracy: Every piece of data must be correct. A single error can have serious consequences. Always double-check your work.

2. Consistency: Use the same format for all similar data. If you write dates as DD/MM/YYYY, use that format for every date. If you write names in Title Case, use Title Case for every name.

3. Confidentiality: Data often contains sensitive information — customer names, phone numbers, financial records. Never share data with unauthorised people. Never leave data visible on your screen when you step away.

4. Efficiency: Accuracy is essential, but speed also matters. Learn keyboard shortcuts. Develop a rhythm. With practice, you can be both fast and accurate.

5. Documentation: When you clean or modify data, document what you changed and why. This creates a record that others can follow."

Topic 5: Typing Accuracy (5 minutes)

Instructor Script: "Typing speed and accuracy are the foundation of data entry. You cannot be a good data entry professional if you type slowly or make many errors.

What Is a Good Typing Speed?

- Minimum for data entry: 25 WPM (words per minute)
- Good: 35–45 WPM

- Excellent: 50+ WPM

What Is a Good Accuracy?

- Minimum for data entry: 95%
- Good: 98%
- Excellent: 99–100%

How to Improve:

- Practise for 15 minutes every day.
- Use a typing practice website.
- Focus on accuracy first — speed will follow.
- Do not look at the keyboard."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Entering Different Data Types (10 minutes)

What the Instructor Prepares:

- Excel open on the projector with a blank spreadsheet

What the Instructor Says and Does: "I am going to enter different types of data into Excel. Watch how I format each type."

1. "I type a name in cell A1: 'Adaeze Okafor.' I use Title Case — first letter of each word capitalised."
2. "I type a phone number in cell A2: '08012345678.' I do not add spaces or dashes — just the digits."
3. "I type a date in cell A3: '01/10/2026.' I use DD/MM/YYYY format."
4. "I type a price in cell A4: '15000.' I do not type the ₦ symbol — I will format the cell as currency."
5. "I select cell A4. I go to the Home tab. I click the Number Format dropdown. I choose 'Currency' and then '₦ English (Nigeria).' The cell now shows ₦15,000."
6. "I type a quantity in cell A5: '25.' This is a whole number — no decimal places needed."

What Learners Should Observe: The instructor entering different data types with correct formatting.

Expected Result: Learners can describe the correct format for each data type.

Demonstration 2: Common Data Entry Errors (5 minutes)

What the Instructor Prepares:

- Excel open on the projector with intentional errors

What the Instructor Says and Does: "Now let me show you some common data entry errors."

1. "Inconsistent date format: '01/10/2026' in one cell and 'October 1, 2026' in another. The fix: use the same format for all dates."
2. "Inconsistent name format: 'adaeze okafor' in one cell and 'ADAEZE OKAFOR' in another. The fix: use Title Case for all names."
3. "Currency symbol typed manually: '₦15,000' as text instead of a number. The fix: type the number and apply currency formatting."
4. "Extra spaces: ' Adaeze ' with spaces before and after. The fix: remove extra spaces."
5. "Wrong decimal places: '15000.00' instead of '15000.' The fix: use the correct number of decimal places."

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "My First Data Entry"

Practical Objective: The learner will enter different types of data into Excel with correct formatting.

Starting Materials:

- Excel open on each machine with a blank spreadsheet

Instructions:

1. "Type the following data into Excel:
    - A1: Your full name (Title Case)
    - A2: Your phone number (digits only, no spaces)
    - A3: Today's date (DD/MM/YYYY format)
    - A4: A price (e.g., 15000 — no currency symbol)
    - A5: A quantity (e.g., 25 — whole number)"
2. "Format cell A4 as currency (₦):
    - Select cell A4.
    - Go to the Home tab.
    - Click the Number Format dropdown.
    - Choose 'Currency' and then '₦ English (Nigeria).'"
3. "Format cell A3 as a date:
    - Select cell A3.
    - Go to the Home tab.
    - Click the Number Format dropdown.
    - Choose 'Short Date.'"
4. "Check your work:
    - Is your name in Title Case?
    - Is your phone number digits only?
    - Is your date in DD/MM/YYYY format?
    - Does cell A4 show the ₦ symbol?
    - Is cell A5 a whole number?"
5. "Save the file as 'My First Data Entry' on the Desktop."

Instructor Prompts:

- "Why did we not type the ₦ symbol manually?" (Answer: Because typing it as text prevents calculations. Formatting the cell as currency keeps it as a number.)
- "Why is consistency important?" (Answer: Inconsistent data is hard to read, sort and analyse.)

Completion Checklist:

- ☐ Name entered in Title Case
- ☐ Phone number entered as digits only
- ☐ Date entered in DD/MM/YYYY format
- ☐ Price formatted as currency (₦)
- ☐ Quantity entered as whole number
- ☐ File saved as "My First Data Entry"

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "Typing Speed Test"

Task: Take a one-minute typing test and record your results.

Instructions:

1. "Go to the typing test website (provided by the instructor)."
2. "Take a one-minute typing test."

3. "Record your results:
    - Words per minute (WPM): _____
    - Accuracy percentage: _____"
4. "Write your results on a piece of paper."
5. "This is your baseline. You will measure your improvement in Session 6."

Requirements:

- Complete the test
- Record WPM and accuracy
- Understand what the results mean

Time Allowance: 10 minutes

Success Criteria:

- Test completed
- Results recorded
- Learner understands their baseline

Hints Available If the Learner Gets Stuck:

- Hint 1: "WPM means words per minute. If you typed 20 words in one minute, your WPM is 20."
- Hint 2: "Accuracy means how many words you typed correctly. If you typed 20 words but 2 were wrong, your accuracy is 90%."
- Hint 3: "Do not worry about being slow. This is your baseline. You will improve with practice."

Complete Answer Key for the Instructor: There is no single correct answer. The learner's result is their personal baseline. The instructor should record each learner's WPM and accuracy for comparison in Session 6.

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "The Wrong Phone Number"

Situation: A receptionist at a hotel in Port Harcourt enters a guest's phone number into the booking system. She is in a hurry and types one digit wrong. The hotel tries to call the guest to confirm the booking. The call goes to a stranger. The guest arrives at the hotel but has no reservation. The guest is furious and leaves a bad review online.

The Problem: A single wrong digit in a phone number caused the hotel to lose a customer and receive a bad review.

Why It Matters: In data entry, accuracy is not optional — it is essential. Every piece of data you enter has consequences.

Relevant Concept: Professional standards. If the receptionist had double-checked the phone number, the problem would not have occurred.

Worked Solution: The receptionist learns to always double-check phone numbers before saving. She develops a habit of reading the number back to the guest to confirm. The hotel also implements a verification step in their booking process.

Reasoning: Accuracy habits — double-checking, reading back, verifying — prevent costly errors.

Possible Mistake: The receptionist might think that one wrong digit is not a big deal. In reality, in data entry, there is no such thing as a small error.

Correction: Every error matters. Develop the habit of double-checking every piece of data you enter.

Transfer Exercise: "You are entering customer data and you are not sure if a phone number is correct. What do you do?"

Professional Connection: Every office job that involves data requires accuracy. Receptionists, administrators, data entry clerks, accountants — everyone. The habits you learn in this course will serve you in any data-related role.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Inconsistent Formatting

Symptoms: Dates, names and phone numbers are formatted differently throughout the dataset. **Likely Cause:** The learner does not follow a consistent format. **How to Investigate:** Look at the dataset. Are all dates in the same format? Are all names in the same case? **Safe Corrective Steps:** Establish a format standard and apply it consistently. **How to Confirm the Problem Is Resolved:** All data follows the same format. **How to Prevent Recurrence:** Create a format guide and refer to it while entering data.

Mistake 2: Typing Currency Symbols Manually

Symptoms: Currency values are stored as text, not numbers. **Likely Cause:** The learner types the ₦ symbol before the number. **How to Investigate:** Click on a currency cell. Is it left-aligned (text) or right-aligned (number)? **Safe Corrective Steps:** Type the number only and apply currency formatting. **How to Confirm the Problem Is Resolved:** Currency values are stored as numbers with currency formatting. **How to Prevent Recurrence:** Always type numbers first, then apply formatting.

Mistake 3: Not Double-Checking Work

Symptoms: Errors are found after the data is submitted. **Likely Cause:** The learner does not verify their work. **How to Investigate:** Ask the learner, "Did you double-check your work?" **Safe Corrective Steps:** Teach the learner to always read back what they typed and compare it to the source. **How to Confirm the Problem Is Resolved:** The learner double-checks their work. **How to Prevent Recurrence:** Make double-checking a habit — every time, without exception.

Mistake 4: Looking at the Keyboard

Symptoms: The learner types slowly and makes many errors. **Likely Cause:** The learner has not memorised the keyboard layout. **How to Investigate:** Watch the learner type. Are they looking down? **Safe Corrective Steps:** Encourage the learner to practise typing without looking at the keyboard. **How to Confirm the Problem Is Resolved:** The learner types without looking down. **How to Prevent Recurrence:** Daily typing practice for 15 minutes.

Mistake 5: Not Saving Work Regularly

Symptoms: The learner loses work when the computer crashes. **Likely Cause:** The learner does not save regularly. **How to Investigate:** Ask the learner, "When did you last save?" **Safe Corrective Steps:** Teach the learner to press Ctrl+S every few minutes. **How to Confirm the Problem Is Resolved:** The learner saves regularly. **How to Prevent Recurrence:** Set a timer that goes off every 5 minutes. When it rings, everyone presses Ctrl+S.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Data Types Reference"

Purpose: To reinforce understanding of data types and their correct formatting.

Realistic Scenario: "You are a data entry professional. Create a reference guide that you can use when entering different types of data. This guide will help you maintain consistency."

Detailed Instructions:

1. Open Microsoft Word or Notepad.
2. Create a reference guide that includes:
    - The four data types (text, numbers, dates, currency)
    - The correct format for each type
    - An example of each type
    - Common errors to avoid for each type
3. Save the file as "My Data Types Reference" on the Desktop.
4. Show the saved file to the instructor before leaving.

Required Deliverables:

- A document named "My Data Types Reference" saved on the Desktop
- The document contains: four data types, correct formats, examples, common errors

Submission Format:

- The file must be saved on the Desktop of the academy machine
- The learner must show the instructor the saved file before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Four data types	All four described accurately	Three described	Two or fewer
Correct formats	All formats correct	Mostly correct	Incorrect
Examples	Clear, accurate examples for all types	Examples for most types	Missing or incorrect
Common errors	Specific, accurate errors listed	Mostly correct	Vague or missing
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved

Total: 25 marks

Common Mistakes to Avoid:

- Writing vague descriptions (e.g., "text is words") without specifics
- Not including examples
- Not saving the file

Instructor Reference Solution:

Text Data:

- Description: Any combination of letters, numbers and symbols
- Correct format: Title Case for names (e.g., "Adaeze Okafor")
- Example: "Adaeze Okafor", "123 Aba Road, Port Harcourt"
- Common errors: Inconsistent capitalisation, misspellings, extra spaces

Numerical Data:

- Description: Numbers used for calculations
- Correct format: No currency symbol, correct decimal places
- Example: "25" (quantity), "15000" (price)
- Common errors: Typing currency symbols manually, wrong decimal places

Date Data:

- Description: Dates in a consistent format
- Correct format: DD/MM/YYYY (e.g., "01/10/2026")
- Example: "01/10/2026", "15/11/2026"
- Common errors: Inconsistent formats, wrong date order

Currency Data:

- Description: Monetary values
- Correct format: Number with currency formatting (e.g., ₦15,000)
- Example: "₦15,000", "₦2,500.50"
- Common errors: Typing the ₦ symbol manually, wrong decimal places

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M. KNOWLEDGE CHECK (5 minutes)

Questions

1. Multiple Choice: What is data entry? a) The process of analysing data b) The process of inputting information into a computer system c) The process of creating databases d) The process of writing code

Answer: b) The process of inputting information into a computer system

2. Short Answer: What are the four main data types?

Answer: Text, numbers, dates and currency.

3. Multiple Choice: What is the standard date format in Nigeria? a) MM/DD/YYYY b) DD/MM/YYYY c) YYYY/MM/DD d) DD-MM-YYYY

Answer: b) DD/MM/YYYY

4. Explain in Your Own Words: Why is it important to format currency values as numbers rather than typing the ₦ symbol manually?

Answer: Typing the ₦ symbol manually stores the value as text, not a number. Text values cannot be used in calculations. By typing the number and applying currency formatting, the value remains a number that can be calculated.

5. Practical Identification: Open Excel. Enter the following data with correct formatting:

- A name in Title Case
- A phone number (digits only)
- A date in DD/MM/YYYY format
- A price formatted as ₦ currency

Answer: The learner correctly enters all four data types with proper formatting.

6. Scenario-Based: Your friend says, "I typed ₦15,000 into Excel but I cannot add it to other numbers. What did I do wrong?" What advice would you give?

Answer: You probably typed the ₦ symbol manually, which stores the value as text. Delete the cell, type just the number 15000, and then apply currency formatting from the Home tab.

7. Error-Detecting: A learner writes in their assignment: "The four data types are text, numbers, dates and percentages." What is wrong with this, and what should they write instead?

Answer: Percentages are not a separate data type — they are a type of numerical data. The four data types are text, numbers, dates and currency.

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you learned the fundamentals of data entry. You learned what data entry is, the four main data types (text, numbers, dates and currency), professional standards for accuracy and consistency, and how to measure your typing speed. You also entered your first data into Excel with correct formatting.

Five Important Takeaways

1. Data entry is the process of inputting information into a computer system.
2. The four data types are text, numbers, dates and currency.
3. Accuracy and consistency are the most important professional standards.
4. Always type numbers first, then apply formatting (especially for currency).
5. Double-check every piece of data you enter.

Questions That Help You Recognise What You Can Now Do

- Can you define data entry?
- Can you name the four data types?
- Can you enter data into Excel with correct formatting?
- Can you explain why consistency matters?
- Do you know your typing speed?

Connection to the Next Session

In the next session, you will learn spreadsheet data entry — how to enter data efficiently into Excel, use keyboard shortcuts and maintain consistent formatting. You now have the foundation. Next, you will learn to work like a professional data entry clerk.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner defines data entry correctly.
- The learner identifies the four data types.
- The learner enters data into Excel with correct formatting.
- The learner explains the importance of accuracy and consistency.
- The learner's typing speed is recorded.

Common Misconceptions

- **"Data entry is just typing."** Some learners think data entry is simply typing fast. Correct this by explaining that accuracy, consistency and understanding data types are equally important.
- **"Formatting does not matter."** Some learners think that as long as the data is there, formatting is optional. Correct this by showing how inconsistent formatting makes data hard to use.
- **"I can type fast enough."** Some learners overestimate their typing speed. Correct this by measuring their WPM and showing them the data entry standard.

Suggested Remedial Exercises

- For learners who cannot identify data types: Provide additional examples and have them classify data into the four types.
- For learners who format currency incorrectly: Have them practise entering currency values and applying formatting multiple times.
- For learners with slow typing speed: Provide a daily typing practice plan and measure their progress weekly.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Provide a "Data Types Reference" handout for reference.
- Pair with a more confident learner for practical exercises.
- Allow extra time for the independent exercise.
- Focus on data types and formatting before typing speed.

Level 2 — Independent (standard):

- Complete all exercises as written.
- Answer all knowledge check questions.
- Complete the assignment to the standard shown in the rubric.

Level 3 — Challenge (for learners who finish early):

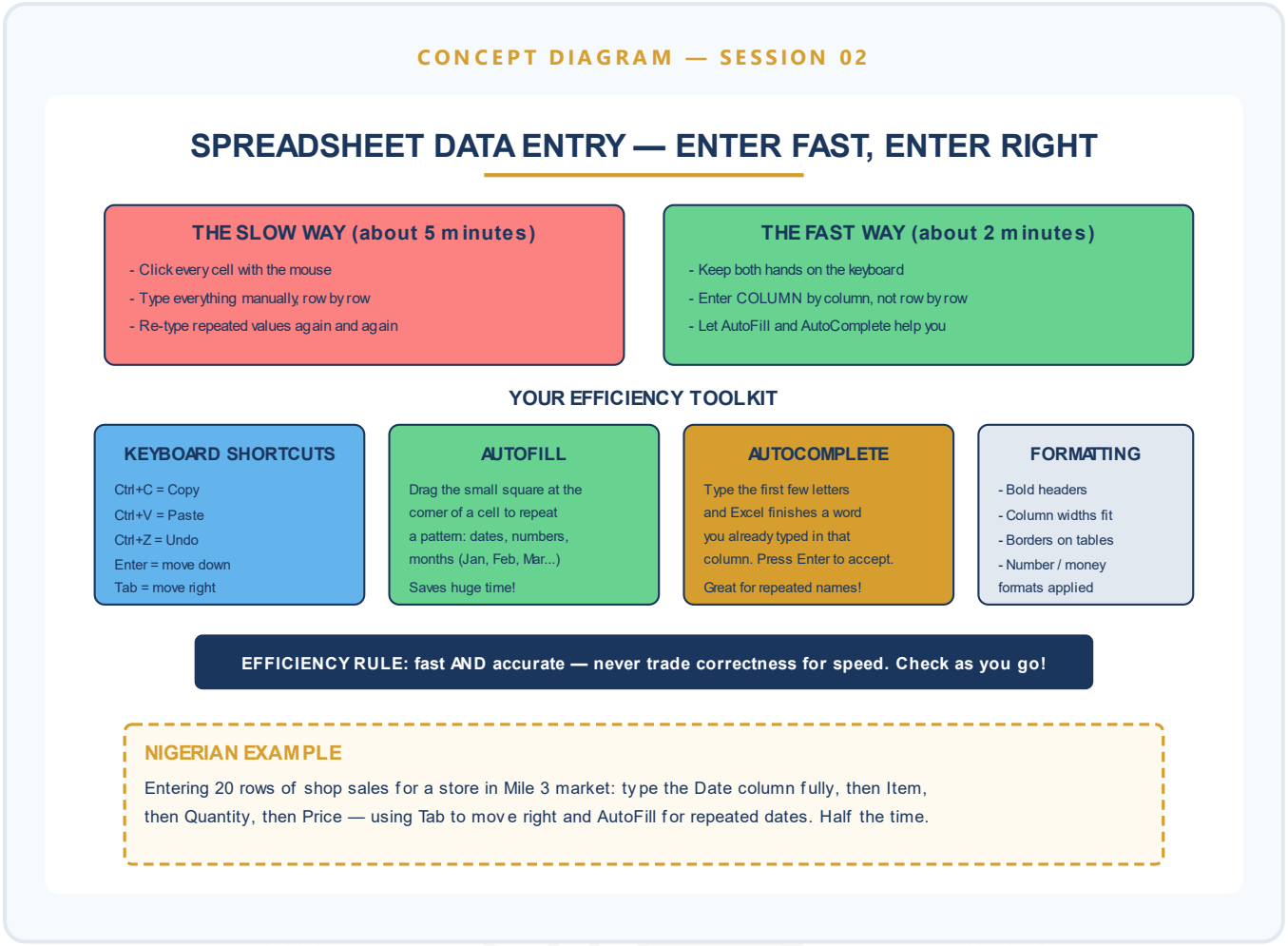
- Ask the learner to create a more detailed data types reference with additional examples.
- Ask the learner to research and write down three more data entry best practices.
- Ask the learner to teach the four data types to a partner.

End of Session 1

Next Session: Session 2 — Spreadsheet Data Entry **What to Bring:** Attention to detail, willingness to practise **Preparation:** Prepare sample sales dataset for data entry practice

SESSION 02 OF 6

Spreadsheet Data Entry



SESSION 2: SPREADSHEET DATA ENTRY

Week: 1 **Module:** Data Entry Fundamentals **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Session 1 — Introduction to Data Entry **Connection to Previous Session:** Builds on Session 1 by teaching efficient spreadsheet data entry **Preparation Required Before Class:**

- Prepare a sample sales dataset (20 rows) for data entry practice
- Ensure Microsoft Excel is installed and functional on all machines

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A. SESSION IDENTIFICATION

Field	Detail
Session Number	2
Title	Spreadsheet Data Entry
Week	1

Field	Detail
Module	Data Entry Fundamentals
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Session 1 — Introduction to Data Entry
Connection to Previous Session	Builds on Session 1
Preparation	Sample sales dataset prepared

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Enter data efficiently** into Excel using best practices.
2. **Use keyboard shortcuts** to improve data entry speed.
3. **Apply consistent formatting** — fonts, alignment, number formats.
4. **Use AutoFill and AutoComplete** to speed up repetitive data entry.
5. **Format an entire dataset** consistently — column widths, header styling, borders.
6. **Explain why** efficiency matters in professional data entry.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- One keyboard per learner
- One mouse per learner
- One monitor per learner
- Projector for instructor demonstration

Software:

- Microsoft Excel 2019 or Microsoft 365

Printed Materials:

- "Excel Keyboard Shortcuts" handout (one per learner)

Sample Files:

- A raw sales dataset (20 rows) for data entry practice (provided by instructor)

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"The Speed Challenge"

Instructor Setup: Display a dataset that needs to be entered into Excel. Time how long it takes to enter it manually vs using shortcuts.

Instructor Script: "Good morning, everyone. Last session, you learned the fundamentals of data entry. Today, we focus on efficiency — how to enter data quickly without sacrificing accuracy.

Watch this. I am going to enter this dataset into Excel. [Enter the data using keyboard shortcuts, AutoFill and AutoComplete.] How long did that take? [Pause.] About 2 minutes. Now watch what happens when I do it the slow way — clicking on each cell with the mouse, typing everything manually. [Enter the same data the slow way.] That took 5 minutes.

The difference? Keyboard shortcuts and Excel's built-in tools. Today, you will learn these tools. By the end of this session, you will be able to enter data twice as fast as you could before."

Expected Learner Response: Recognition that efficiency tools save significant time. Curiosity about the shortcuts.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, quick review of last session."

Review Questions and Expected Answers:

1. "What are the four main data types?"
   - Text, numbers, dates and currency.
2. "What is the standard date format in Nigeria?"
   - DD/MM/YYYY.
3. "Why should you not type the ₦ symbol manually?"
   - Because it stores the value as text, not a number, preventing calculations.
4. "What is the most important professional standard in data entry?"
   - Accuracy.
5. "What was your typing speed in the last session?"
   - (Each learner states their WPM.)

F. FULL LECTURE NOTES (35 minutes)

Topic 1: Data Entry Best Practices (10 minutes)

Instructor Script: "Here are the best practices for efficient, accurate data entry in Excel.

1. Plan Before You Enter:

- Look at the source data before you start.
- Identify the columns you need.
- Decide on the format for each column before you begin.

2. Enter Data Column by Column:

- Instead of entering one row at a time, enter one column at a time.
- This is faster because you can use AutoFill for repetitive data.
- It also reduces errors because you focus on one data type at a time.

3. Use Tab to Move Right, Enter to Move Down:

- After typing in a cell, press Tab to move to the next cell to the right.
- After typing in the last cell of a row, press Enter to move to the first cell of the next row.
- This keeps your hands on the keyboard and eliminates mouse clicking.

4. Double-Check as You Go:

- After entering each column, scan it for errors.
- It is easier to fix errors immediately than to find them later.

5. Save Regularly:

- Press Ctrl+S every few minutes.

- Do not wait until the end to save."

Topic 2: Keyboard Shortcuts for Efficiency (10 minutes)

Instructor Script: "Keyboard shortcuts save time because you do not have to move your hand to the mouse. Here are the most important shortcuts for data entry.

Navigation Shortcuts:

- **Tab:** Move to the next cell to the right.
- **Enter:** Move to the cell below.
- **Shift+Tab:** Move to the cell to the left.
- **Shift+Enter:** Move to the cell above.
- **Ctrl+Home:** Move to cell A1.
- **Ctrl+End:** Move to the last cell with data.
- **Ctrl+Arrow Keys:** Move to the edge of the data range.

Editing Shortcuts:

- **Ctrl+C:** Copy.
- **Ctrl+V:** Paste.
- **Ctrl+X:** Cut.
- **Ctrl+Z:** Undo.
- **Ctrl+Y:** Redo.
- **Delete:** Clear the contents of a cell.
- **F2:** Edit the active cell.

Selection Shortcuts:

- **Ctrl+A:** Select all.
- **Shift+Arrow Keys:** Extend selection.
- **Ctrl+Shift+Arrow Keys:** Select to the edge of the data range.

Formatting Shortcuts:

- **Ctrl+B:** Bold.
- **Ctrl+I:** Italic.
- **Ctrl+U:** Underline.
- **Ctrl+1:** Open the Format Cells dialog."

Topic 3: AutoFill and AutoComplete (5 minutes)

Instructor Script: "Excel has two features that speed up repetitive data entry.

AutoFill: If you type 'January' in a cell and drag the fill handle (the small square at the bottom-right corner of the cell), Excel automatically fills in February, March, April, and so on.

You can also AutoFill numbers, dates and custom lists.

AutoComplete: If you type the first few letters of text that already exists in the column, Excel suggests the rest. Press Enter to accept the suggestion.

Example: If 'Adaeze' is already in the column and you type 'Ad', Excel suggests 'Adaeze'. Press Enter to accept."

Topic 4: Consistent Formatting (10 minutes)

Instructor Script: "Consistent formatting makes data professional and easy to read. Here is how to format an entire dataset consistently.

1. Header Row:

- Make the header row bold.

- Apply a background colour.
- Freeze the header row so it stays visible when you scroll.

2. Column Widths:

- Adjust column widths so all data is visible.
- Double-click the column border to auto-fit.

3. Number Formats:

- Apply the correct number format to each column (currency, date, number, text).

4. Alignment:

- Left-align text columns.
- Right-align number columns.
- Centre-align headers.

5. Borders:

- Add borders to separate data visually.
- Use thin borders for a clean look.

How to Apply Formatting to an Entire Column:

1. Click on the column header (letter) to select the entire column.
2. Apply the formatting (bold, number format, alignment, etc.).
3. The formatting applies to all cells in the column."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Entering a Dataset Efficiently (10 minutes)

What the Instructor Prepares:

- A raw sales dataset (20 rows) projected on the screen
- Excel open on the projector

What the Instructor Says and Does: "I am going to enter this sales dataset efficiently. Watch my technique."

1. "I look at the source data and identify the columns: Date, Customer Name, Product, Quantity, Unit Price, Total."
2. "I enter the headers in row 1."
3. "I format the header row: bold, background colour, centred."
4. "I enter the Date column first, using AutoFill for repetitive dates."
5. "I enter the Customer Name column, using AutoComplete for repeated names."
6. "I enter the Product column."
7. "I enter the Quantity column."
8. "I enter the Unit Price column."
9. "I enter the Total column using a formula: =Quantity*Unit Price."
10. "I format each column: dates as DD/MM/YYYY, prices as ₦ currency, quantities as whole numbers."
11. "I adjust column widths."
12. "I add borders."
13. "I freeze the header row."
14. "I save the file."

What Learners Should Observe: The instructor entering data efficiently using shortcuts, AutoFill and consistent formatting.

Expected Result: Learners can describe the efficient data entry technique.

Demonstration 2: Using Keyboard Shortcuts (5 minutes)

What the Instructor Prepares:

- Excel open on the projector

What the Instructor Says and Does: "Now I will demonstrate the keyboard shortcuts."

1. "I type 'Test' in cell A1. I press Tab. The cursor moves to B1."
2. "I type 'Data'. I press Enter. The cursor moves to A2."
3. "I press Ctrl+Home. The cursor moves to A1."
4. "I press Ctrl+Shift+End. All data is selected."
5. "I press Ctrl+B. The selected text becomes bold."
6. "I press Ctrl+Z. The bold is undone."
7. "I press Ctrl+S. The file is saved."

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "Enter the Sales Dataset"

Practical Objective: The learner will enter a 20-row sales dataset into Excel efficiently with consistent formatting.

Starting Materials:

- A raw sales dataset (provided by instructor — printed or projected)
- Excel open on each machine with a blank spreadsheet

Instructions:

1. "Look at the source data. Identify the columns you need."
2. "Enter the headers in row 1: Date, Customer Name, Product, Quantity, Unit Price, Total."
3. "Format the header row: bold, background colour, centred."
4. "Enter the data column by column:
    - Date column first (use AutoFill for repetitive dates)
    - Customer Name column (use AutoComplete for repeated names)
    - Product column
    - Quantity column
    - Unit Price column
    - Total column (use formula: =Quantity*Unit Price)"
5. "Format each column:
    - Dates: DD/MM/YYYY
    - Names: Title Case
    - Prices: ₦ currency
    - Quantities: whole numbers"
6. "Adjust column widths so all data is visible."
7. "Add borders to all data."
8. "Freeze the header row."
9. "Save the file as 'Sales Dataset' on the Desktop."

Instructor Prompts:

- "Are you entering data column by column or row by row?" (Answer: Column by column.)
- "Are you using Tab and Enter to navigate?" (Answer: Yes.)
- "Is your formatting consistent across all columns?" (Answer: Yes.)

Completion Checklist:

- ☐ Headers entered and formatted
- ☐ All 20 rows of data entered

- ☐ Dates in DD/MM/YYYY format
- ☐ Names in Title Case
- ☐ Prices formatted as ₦ currency
- ☐ Quantities as whole numbers
- ☐ Totals calculated using formulas
- ☐ Column widths adjusted
- ☐ Borders added
- ☐ Header row frozen
- ☐ File saved as "Sales Dataset"

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "Shortcut Speed Test"

Scenario: "You are a data entry professional. Your supervisor asks you to enter a small dataset as quickly as possible. Use only keyboard shortcuts — no mouse."

Task: Enter the following data into Excel using only keyboard shortcuts:

Name	Department	Salary
Adaeze Okafor	Sales	85000
Chinedu Obi	IT	120000
Funmi Adewale	HR	95000
Tunde Bakare	Finance	110000
Ngozi Eze	Marketing	90000

Requirements:

- Use Tab to move right, Enter to move down
- Use Ctrl+B to bold the headers
- Use Ctrl+1 to format the Salary column as currency
- Use Ctrl+S to save

Time Allowance: 10 minutes

Success Criteria:

- All data entered correctly
- Headers bold
- Salary formatted as currency
- File saved
- No mouse used for data entry

Hints Available If the Learner Gets Stuck:

- Hint 1: "Press Tab to move right. Press Enter to move down."
- Hint 2: "Press Ctrl+B to bold. Press Ctrl+1 to open the Format Cells dialog."
- Hint 3: "Press Ctrl+S to save."

Complete Answer Key for the Instructor: The learner should enter all 5 rows of data correctly, bold the headers, format the Salary column as currency and save the file — all using keyboard shortcuts only.

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "The Efficient Data Entry Clerk"

Situation: Two data entry clerks work in the same office. They have the same amount of data to enter. Clerk A uses the mouse for everything — clicking on each cell, typing manually, formatting one cell at a time. Clerk B uses keyboard shortcuts, AutoFill and column-by-column entry.

At the end of the day, Clerk B has entered twice as much data as Clerk A — with fewer errors.

The Problem: Clerk A is slow and error-prone because they do not use efficiency tools.

Why It Matters: In a professional workplace, productivity matters. Employers value employees who can work efficiently without sacrificing accuracy.

Relevant Concept: Efficiency tools. If Clerk A learned the shortcuts and techniques taught in this session, they would be twice as productive.

Worked Solution: Clerk A learns keyboard shortcuts, AutoFill and column-by-column entry. Within a week, Clerk A is as productive as Clerk B.

Reasoning: Efficiency is not about working harder — it is about working smarter. The right tools and techniques make you more productive.

Possible Mistake: Clerk A might think that using the mouse is just as fast. In reality, moving your hand between the keyboard and mouse hundreds of times per hour adds up to significant wasted time.

Correction: Keep your hands on the keyboard. Use shortcuts for everything. The mouse is for tasks that cannot be done with the keyboard.

Transfer Exercise: "You are entering data and you keep reaching for the mouse. What are three things you can do with keyboard shortcuts instead?"

Professional Connection: Every data entry job requires efficiency. The faster and more accurately you can enter data, the more valuable you are to an employer.

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Entering Data Row by Row

Symptoms: The learner enters one row at a time, switching between data types. **Likely Cause:** The learner does not know the column-by-column technique. **How to Investigate:** Watch the learner enter data. Are they switching between data types? **Safe Corrective Steps:** Teach the learner to enter one column at a time. **How to Confirm the Problem Is Resolved:** The learner enters data column by column. **How to Prevent Recurrence:** Remind the learner: "Column by column, not row by row."

Mistake 2: Using the Mouse for Navigation

Symptoms: The learner clicks on each cell with the mouse before typing. **Likely Cause:** The learner does not know the keyboard navigation shortcuts. **How to Investigate:** Watch the learner's hands. Are they reaching for the mouse? **Safe Corrective Steps:** Teach the learner to use Tab and Enter for navigation. **How to Confirm the Problem Is Resolved:** The learner uses Tab and Enter. **How to Prevent Recurrence:** Remind the learner: "Keyboard for navigation, mouse only when necessary."

Mistake 3: Inconsistent Formatting

Symptoms: Some cells are formatted differently from others in the same column. **Likely Cause:** The learner formats cells individually instead of formatting entire columns. **How to Investigate:** Look at the dataset. Is the formatting consistent within each column? **Safe Corrective Steps:** Teach the learner to select entire columns and apply formatting. **How to Confirm the Problem Is Resolved:** Formatting is consistent within each column. **How to Prevent Recurrence:** Remind the learner: "Format the column, not the cell."

Mistake 4: Not Using AutoFill

Symptoms: The learner types repetitive data manually. **Likely Cause:** The learner does not know about AutoFill. **How to Investigate:** Ask the learner, "Did you use AutoFill for repetitive data?" **Safe Corrective Steps:** Demonstrate AutoFill and have the learner practise. **How to Confirm the Problem Is Resolved:** The learner uses AutoFill. **How to Prevent Recurrence:** Remind the learner: "If you are typing the same thing more than twice, use AutoFill."

Mistake 5: Not Freezing the Header Row

Symptoms: The header row scrolls out of view when scrolling through data. **Likely Cause:** The learner does not know about freezing panes. **How to Investigate:** Ask the learner, "Can you see the header row when you scroll down?" **Safe Corrective Steps:** Teach the learner to freeze the header row. **How to Confirm the Problem Is Resolved:** The header row stays visible when scrolling. **How to Prevent Recurrence:** Remind the learner: "Always freeze the header row."

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Sales Dataset"

Purpose: To reinforce efficient data entry skills by having the learner enter a complete sales dataset.

Realistic Scenario: "You are a data entry clerk at a small business. Your supervisor gives you a list of 20 sales transactions and asks you to enter them into Excel. The dataset must be accurate, consistently formatted and saved professionally."

Detailed Instructions:

1. Open the "Raw Sales Data" file provided by the instructor.
2. Enter all 20 rows of data into a new Excel spreadsheet.
3. Use efficient data entry techniques:
    - Enter data column by column
    - Use Tab and Enter for navigation
    - Use AutoFill for repetitive data
    - Use AutoComplete for repeated names
4. Apply consistent formatting:
    - Header row: bold, background colour, centred
    - Dates: DD/MM/YYYY
    - Names: Title Case
    - Prices: ₦ currency
    - Quantities: whole numbers
5. Calculate the Total column using formulas.
6. Adjust column widths.
7. Add borders.
8. Freeze the header row.
9. Save the file as "My Sales Dataset" on the Desktop.
10. Show the saved file to the instructor before leaving.

Required Deliverables:

- An Excel workbook named "My Sales Dataset" saved on the Desktop
- The workbook contains: 20 rows of data, consistent formatting, formulas for totals, frozen header row

Submission Format:

- The file must be saved on the Desktop of the academy machine
- The learner must show the instructor the saved file before leaving

Deadline Recommendation:

- Before leaving the session (in-class assignment)

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Data accuracy	All 20 rows correct	18–19 correct	17 or fewer correct
Efficiency techniques	All techniques used	Most used	Few used

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Header formatting	Bold, coloured, centred	Mostly correct	Incorrect
Date format	All DD/MM/YYYY	Mostly correct	Inconsistent
Currency format	All ₦ formatted	Mostly correct	Inconsistent
Formulas	All totals calculated	Most calculated	Few calculated
Column widths	All adjusted	Most adjusted	Not adjusted
Borders	All data bordered	Most bordered	No borders
Header frozen	Frozen	Not frozen	Not applicable
File saved correctly	Saved on Desktop with correct name	Saved but wrong location or name	Not saved

Total: 50 marks

Common Mistakes to Avoid:

- Entering data row by row instead of column by column
- Using the mouse for navigation instead of keyboard shortcuts
- Not using AutoFill for repetitive data
- Inconsistent formatting within columns
- Not freezing the header row
- Not saving the file

Instructor Reference Solution:

The final workbook should have:

- 20 rows of accurate data
- Headers: bold, background colour, centred
- Dates: DD/MM/YYYY format
- Names: Title Case
- Prices: ₦ currency format
- Quantities: whole numbers
- Totals: calculated using formulas (=Quantity*Unit Price)
- Column widths: adjusted to fit all data
- Borders: applied to all data
- Header row: frozen
- Saved as "My Sales Dataset.xlsx" on the Desktop

M. KNOWLEDGE CHECK (5 minutes)

Questions

1. Multiple Choice: What is the most efficient way to enter data into Excel? a) Row by row, using the mouse b) Column by column, using keyboard shortcuts c) Row by row, using keyboard shortcuts d) Column by column, using the mouse

Answer: b) Column by column, using keyboard shortcuts

2. Short Answer: What are three keyboard shortcuts that improve data entry efficiency?

Answer: Tab (move right), Enter (move down), Ctrl+S (save). Other acceptable answers: Ctrl+C, Ctrl+V, Ctrl+Z, Ctrl+B, Ctrl+1, Ctrl+Home, Ctrl+End.

3. Multiple Choice: What does AutoFill do? a) Automatically formats cells b) Automatically fills in repetitive data based on a pattern c) Automatically saves the file d) Automatically checks for errors

Answer: b) Automatically fills in repetitive data based on a pattern

4. Explain in Your Own Words: Why should you format entire columns instead of individual cells?

Answer: Formatting entire columns ensures consistency across all data in that column. It is also faster than formatting cells one by one.

5. Practical Identification: Open Excel. Demonstrate the following shortcuts: Tab, Enter, Ctrl+B, Ctrl+S, Ctrl+Z.

Answer: The learner correctly demonstrates all five shortcuts.

6. Scenario-Based: Your friend says, "I entered all my data but the header row scrolls away when I scroll down. How do I fix this?" What advice would you give?

Answer: Freeze the header row. Select the row below the header, go to the View tab, click "Freeze Panes" and choose "Freeze Top Row."

7. Error-Detecting: A learner writes in their assignment: "I entered the data row by row because it is faster." What is wrong with this, and what should they write instead?

Answer: Entering data row by row is slower because you switch between data types and cannot use AutoFill effectively. The learner should write: "I entered the data column by column because it is faster and allows me to use AutoFill for repetitive data."

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you learned efficient spreadsheet data entry. You learned best practices, keyboard shortcuts, AutoFill, AutoComplete and how to format an entire dataset consistently. You also entered a 20-row sales dataset using these techniques.

Five Important Takeaways

1. Enter data column by column, not row by row.
2. Use keyboard shortcuts — Tab, Enter, Ctrl+S — instead of the mouse.
3. Use AutoFill and AutoComplete for repetitive data.
4. Format entire columns, not individual cells.
5. Always freeze the header row and save regularly.

Questions That Help You Recognise What You Can Now Do

- Can you enter data efficiently using keyboard shortcuts?
- Can you use AutoFill for repetitive data?
- Can you format an entire dataset consistently?
- Can you calculate totals using formulas?
- Can you freeze the header row?

Connection to the Next Session

In the next session, you will learn data quality and cleaning — how to identify and correct errors, standardise formats and document your changes. You now know how to enter data efficiently. Next, you will learn to make it perfect.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner enters data column by column.
- The learner uses keyboard shortcuts for navigation.
- The learner uses AutoFill and AutoComplete.
- The learner applies consistent formatting to entire columns.
- The learner freezes the header row and saves regularly.

Common Misconceptions

- **"Row by row is just as fast."** Some learners think the order does not matter. Correct this by demonstrating the time difference.
- **"The mouse is more precise."** Some learners think clicking is more accurate. Correct this by showing that keyboard navigation is both faster and more accurate.
- **"Formatting individual cells is fine."** Some learners format cells one at a time. Correct this by showing how column formatting ensures consistency.

Suggested Remedial Exercises

- For learners who use the mouse for navigation: Have them practise entering data using only the keyboard for 10 minutes.
- For learners who do not use AutoFill: Provide a dataset with repetitive data and require them to use AutoFill.
- For learners who format cells individually: Have them reformat an entire dataset by selecting columns and applying formatting.

Differentiation Guidance

Level 1 — Supported (for learners who struggle):

- Provide a partially completed dataset for them to finish.
- Pair with a more confident learner for practical exercises.
- Allow extra time for the independent exercise.
- Focus on basic shortcuts (Tab, Enter, Ctrl+S) before advanced ones.

Level 2 — Independent (standard):

- Complete all exercises as written.
- Answer all knowledge check questions.
- Complete the assignment to the standard shown in the rubric.

Level 3 — Challenge (for learners who finish early):

- Ask the learner to enter a larger dataset (50+ rows) using the same techniques.
- Ask the learner to research and write down three more Excel shortcuts.
- Ask the learner to create a custom AutoFill list.

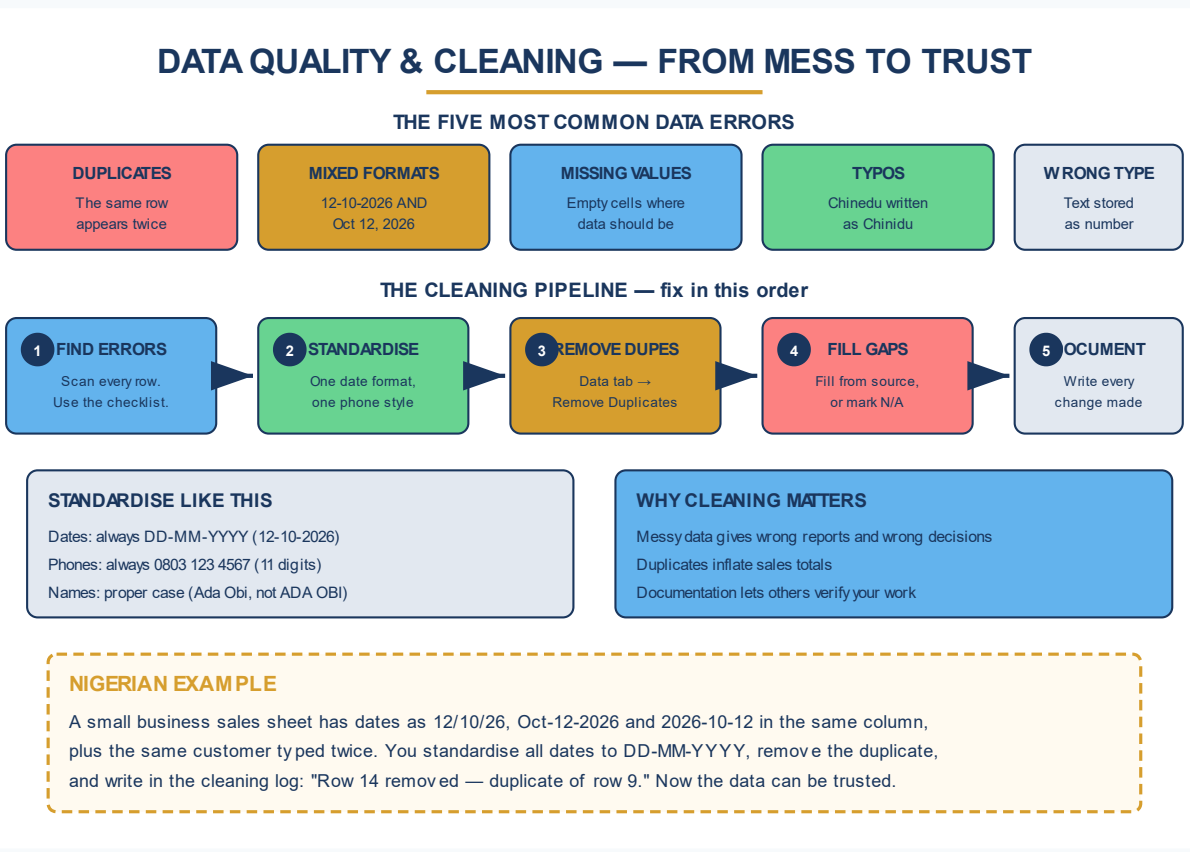
End of Session 2

Next Session: Session 3 — Data Quality and Cleaning **What to Bring:** Attention to detail, patience, willingness to find and fix errors **Preparation:** Prepare messy dataset for cleaning exercise

SESSION 03 OF 6

Data Quality and Cleaning

CONCEPT DIAGRAM — SESSION 03



SESSION 3: DATA QUALITY AND CLEANING

Week: 1 **Module:** Data Entry Fundamentals **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Sessions 1–2 — Introduction to Data Entry, Spreadsheet Data Entry **Connection to Previous Session:** Builds on Session 2 by teaching data quality and cleaning **Preparation Required Before Class:**

- Prepare a messy dataset with intentional errors (inconsistent formats, duplicates, missing values, typos)
- Ensure Microsoft Excel is installed and functional on all machines

A. SESSION IDENTIFICATION

Field	Detail
Session Number	3
Title	Data Quality and Cleaning
Week	1

Field	Detail
Module	Data Entry Fundamentals
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Sessions 1–2
Connection to Previous Session	Builds on Session 2
Preparation	Messy dataset prepared

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Identify common data errors** — duplicates, inconsistencies, missing values, typos, formatting issues.
2. **Standardise formats** — dates, phone numbers, names, currency.
3. **Remove duplicates** using Excel's built-in tools.
4. **Fill missing values** appropriately.
5. **Document all changes** made during the cleaning process.
6. **Explain why** data cleaning is essential for data quality.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- Projector for instructor demonstration

Software:

- Microsoft Excel 2019 or Microsoft 365

Printed Materials:

- "Data Cleaning Checklist" (one per learner)

Sample Files:

- A messy dataset with intentional errors (provided by instructor)

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"The Messy Dataset"

Instructor Setup: Display a messy dataset on the projector — with inconsistent dates, misspelled names, duplicate rows, missing values and mixed formatting.

Instructor Script: "Good morning, everyone. Look at this dataset. It contains sales records for a small business. But it is a mess. The dates are in three different formats. Some names are misspelled. There are duplicate rows. Some cells are empty. The currency formatting is inconsistent.

If you tried to use this dataset to create a report or make a decision, you would get wrong answers. The data is there, but it is not usable in its current state.

Today, you will learn how to clean this dataset — to identify errors, standardise formats, remove duplicates and fill missing values. By the end of this session, you will turn this mess into a clean, professional dataset."

Expected Learner Response: Recognition that messy data is a real problem. Curiosity about how to fix it.

E. PREREQUISITE REVIEW (5 minutes)

Instructor Script: "Before we start, quick review of last session."

Review Questions and Expected Answers:

1. "What is the most efficient way to enter data into Excel?"
    - Column by column, using keyboard shortcuts.
2. "What does AutoFill do?"
    - It automatically fills in repetitive data based on a pattern.
3. "Why should you format entire columns instead of individual cells?"
    - To ensure consistency and save time.
4. "What is the most important professional standard in data entry?"
    - Accuracy.
5. "What are the four main data types?"
    - Text, numbers, dates and currency.

F. FULL LECTURE NOTES (35 minutes)

Topic 1: Common Data Errors (10 minutes)

Instructor Script: "Here are the most common data errors you will encounter:

1. Inconsistent Date Formats:

- '01/10/2026' vs 'October 1, 2026' vs '2026-10-01'
- Fix: Standardise to one format (DD/MM/YYYY in Nigeria).

2. Inconsistent Name Formats:

- 'adaeze okafor' vs 'ADAEZE OKAFOR' vs 'Adaeze Okafor'
- Fix: Standardise to Title Case.

3. Misspellings:

- 'Adaeze' vs 'Adaeeze' vs 'Adaeza'
- Fix: Correct the spelling. Use the source document to verify.

4. Duplicate Records:

- The same transaction entered twice.
- Fix: Identify and remove duplicates.

5. Missing Values:

- Empty cells where data should be.
- Fix: Fill in the missing data from the source document. If the data is not available, mark it as 'Not Available' or 'N/A'.

6. Inconsistent Currency Formatting:

- '₦15,000' vs '15000' vs 'N15,000' vs '15000.00'
- Fix: Standardise to ₦ currency format with commas and two decimal places.

7. Extra Spaces:

- ' Adaeze ' vs 'Adaeze'
- Fix: Remove leading and trailing spaces.

8. Wrong Data Types:

- Text stored as numbers or numbers stored as text.
- Fix: Convert to the correct data type."

Topic 2: Standardising Formats (10 minutes)

Instructor Script: "Standardising means making all data in a column follow the same format. Here is how to standardise each data type:

Dates:

1. Select the date column.
2. Go to the Home tab.
3. Click the Number Format dropdown.
4. Choose 'Short Date' or 'Custom' and enter DD/MM/YYYY.

Names:

1. Use the PROPER function to convert to Title Case: `=PROPER(A1)`
2. Copy the result and paste as values.
3. Delete the original column.

Phone Numbers:

1. Ensure all numbers are digits only (no spaces, dashes or symbols).
2. If needed, use the TEXT function to format: `=TEXT(A1, "00000000000")`

Currency:

1. Select the currency column.
2. Go to the Home tab.
3. Click the Number Format dropdown.
4. Choose 'Currency' and then '₦ English (Nigeria)'."

Topic 3: Removing Duplicates (5 minutes)

Instructor Script: "Excel has a built-in tool to remove duplicate rows.

How to Remove Duplicates:

1. Select the data range.
2. Go to the Data tab.
3. Click 'Remove Duplicates'.
4. Select the columns to check for duplicates.
5. Click OK.
6. Excel removes duplicate rows and tells you how many were removed.

Important Note: Always check the data before removing duplicates. Make sure you are removing true duplicates, not just rows that happen to have the same value in one column."

Topic 4: Filling Missing Values (5 minutes)

Instructor Script: "Missing values are cells that are empty but should contain data.

How to Fill Missing Values:

1. Identify the missing cells (use Ctrl+G > Special > Blanks to find them).
2. Fill in the missing data from the source document.
3. If the data is not available, enter 'N/A' or 'Not Available'.

4. Document what you filled in and what you could not find.

Important Note: Never make up data. If you do not know the correct value, mark it as missing and flag it for follow-up."

Topic 5: Documenting Changes (5 minutes)

Instructor Script: "When you clean data, you must document every change you make. This creates a record that others can follow and verify.

What to Document:

1. What errors you found.
2. What changes you made.
3. How you made each change.
4. What data you could not find or correct.
5. The date you made the changes.

Documentation Format: Create a 'Data Cleaning Log' — a separate sheet in the workbook or a separate document that lists all changes."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Cleaning a Messy Dataset (15 minutes)

What the Instructor Prepares:

- A messy dataset open on the projector

What the Instructor Says and Does: "I am going to clean this messy dataset step by step."

1. "First, I identify the errors:
    - Dates are in three different formats.
    - Names have inconsistent capitalisation.
    - There are two duplicate rows.
    - Three cells are missing values.
    - Currency formatting is inconsistent."
2. "I standardise the dates:
    - I select the date column.
    - I apply DD/MM/YYYY format."
3. "I standardise the names:
    - I use the PROPER function to convert to Title Case.
    - I copy and paste as values."
4. "I remove duplicates:
    - I go to the Data tab.
    - I click 'Remove Duplicates'.
    - I select all columns.
    - Excel removes 2 duplicate rows."
5. "I fill missing values:
    - I use Ctrl+G > Special > Blanks to find empty cells.
    - I fill in the missing data from the source document.
    - I mark one cell as 'N/A' because the data is not available."
6. "I standardise currency:
    - I select the price column.

    - I apply ₦ currency format."
7. "I create a Data Cleaning Log:
    - I create a new sheet called 'Cleaning Log'.
    - I list every change I made."

What Learners Should Observe: The instructor systematically cleaning a messy dataset.

Expected Result: Learners can describe the data cleaning process.

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "Clean the Messy Dataset"

Practical Objective: The learner will clean a messy dataset by identifying errors, standardising formats, removing duplicates and filling missing values.

Starting Materials:

- A messy dataset open on each machine (provided by instructor)
- A printed "Data Cleaning Checklist"

Instructions:

1. "Open the 'Messy Dataset' file."
2. "Identify all errors:
    - Inconsistent date formats
    - Inconsistent name formats
    - Duplicate rows
    - Missing values
    - Inconsistent currency formatting"
3. "Standardise the date column to DD/MM/YYYY."
4. "Standardise the name column to Title Case."
5. "Remove duplicate rows."
6. "Fill missing values (use 'N/A' if the data is not available)."
7. "Standardise the currency column to ₦ format."
8. "Create a 'Cleaning Log' sheet that documents every change you made."
9. "Save the file as 'Cleaned Dataset' on the Desktop."

Instructor Prompts:

- "What errors did you find?" (Answer: Varies — dates, names, duplicates, missing values, currency.)
- "How did you standardise the names?" (Answer: Used the PROPER function.)
- "How many duplicates did you remove?" (Answer: Varies.)
- "What did you do with missing values?" (Answer: Filled from source or marked as N/A.)

Completion Checklist:

- [ ] All errors identified
- [ ] Dates standardised to DD/MM/YYYY
- [ ] Names standardised to Title Case
- [ ] Duplicates removed
- [ ] Missing values filled or marked as N/A
- [ ] Currency standardised to ₦ format
- [ ] Cleaning Log created
- [ ] File saved as "Cleaned Dataset"

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "Error Identification"

Scenario: "You are given a dataset with 10 intentional errors. Your task is to find and describe each error."

Task: Open the "Error Hunt" file provided by the instructor. Find all 10 errors and write down:

1. What the error is
2. Where it is (cell reference)
3. How to fix it

Time Allowance: 10 minutes

Success Criteria:

- All 10 errors found
- Each error described accurately
- Fix suggested for each error

Hints Available If the Learner Gets Stuck:

- Hint 1: "Check for inconsistent date formats."
- Hint 2: "Check for misspelled names."
- Hint 3: "Check for duplicate rows."
- Hint 4: "Check for missing values."
- Hint 5: "Check for inconsistent currency formatting."

Complete Answer Key for the Instructor: The 10 errors should include: 2 inconsistent date formats, 2 misspelled names, 2 duplicate rows, 2 missing values, 1 inconsistent currency format, 1 extra space in a name.

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "The Report That Was Wrong"

Situation: A business owner asks their administrator to create a sales report from the company's dataset. The administrator uses the dataset without cleaning it. The report shows incorrect totals because of duplicate rows and missing values. The business owner makes a decision based on the wrong report and loses money.

The Problem: The dataset was not cleaned before use. Duplicates inflated the totals. Missing values skewed the averages.

Why It Matters: Clean data is the foundation of good decisions. Dirty data leads to wrong conclusions and costly mistakes.

Relevant Concept: Data cleaning. If the administrator had cleaned the dataset first, the report would have been accurate.

Worked Solution: The administrator learns to clean data before using it. They create a cleaning checklist and follow it every time. The next report is accurate and the business owner makes the right decision.

Transfer Exercise: "Your friend is about to create a report from a dataset. What advice would you give them?"

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Not Checking for Duplicates

Symptoms: Totals are inflated because of duplicate rows. **Fix:** Always check for duplicates before using a dataset.

Mistake 2: Making Up Missing Data

Symptoms: The learner fills in missing values with guessed data. **Fix:** Never make up data. Use the source document or mark as N/A.

Mistake 3: Not Documenting Changes

Symptoms: The learner cleans the data but does not record what they changed. **Fix:** Always create a cleaning log.

Mistake 4: Over-Cleaning

Symptoms: The learner removes data that is actually valid. **Fix:** Check with the source document before deleting anything.

Mistake 5: Not Standardising Formats

Symptoms: The learner fixes errors but leaves formats inconsistent. **Fix:** Always standardise formats after fixing errors.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Cleaned Dataset"

Clean a messy dataset and document all changes.

Requirements:

- All errors identified and fixed
- Formats standardised
- Duplicates removed
- Missing values handled appropriately
- Cleaning log created
- File saved as "My Cleaned Dataset"

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Error identification	All errors found	Most found	Few found
Format standardisation	All formats correct	Mostly correct	Inconsistent
Duplicate removal	All duplicates removed	Most removed	Few removed
Missing values	All handled appropriately	Most handled	Few handled
Cleaning log	Complete and detailed	Mostly complete	Incomplete
File saved correctly	Saved on Desktop with correct name	Saved but wrong location	Not saved

Total: 30 marks

M. KNOWLEDGE CHECK (5 minutes)

1. What are the five most common data errors? Answer: Inconsistent formats, misspellings, duplicates, missing values and wrong data types.

2. How do you remove duplicates in Excel? Answer: Select the data, go to the Data tab, click "Remove Duplicates," select the columns and click OK.

3. What should you do if you cannot find a missing value? Answer: Mark it as 'N/A' or 'Not Available' and flag it for follow-up. Never make up data.

4. Why is it important to document data cleaning changes? Answer: It creates a record that others can follow and verify. It also helps you remember what you did.

5. What Excel function converts text to Title Case? Answer: PROPER

N. REFLECTION AND RECAP (5 minutes)

Today you learned data quality and cleaning — how to identify errors, standardise formats, remove duplicates, fill missing values and document changes. In the next session, you will learn data validation — how to check data against source documents.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner identifies common data errors.
- The learner standardises formats correctly.
- The learner removes duplicates.
- The learner handles missing values appropriately.
- The learner documents all changes.

Common Misconceptions

- **"Cleaning is just fixing typos."** Teach: cleaning includes standardising formats, removing duplicates and handling missing values.
- **"I can make up missing data."** Teach: never make up data — mark it as N/A.
- **"Documentation is optional."** Teach: documentation is essential for transparency and verification.

Differentiation Guidance

- Level 1: Provide a partially cleaned dataset and guide them through the remaining steps.
- Level 2: Complete the exercise as written.
- Level 3: Create a more complex cleaning challenge with additional error types.

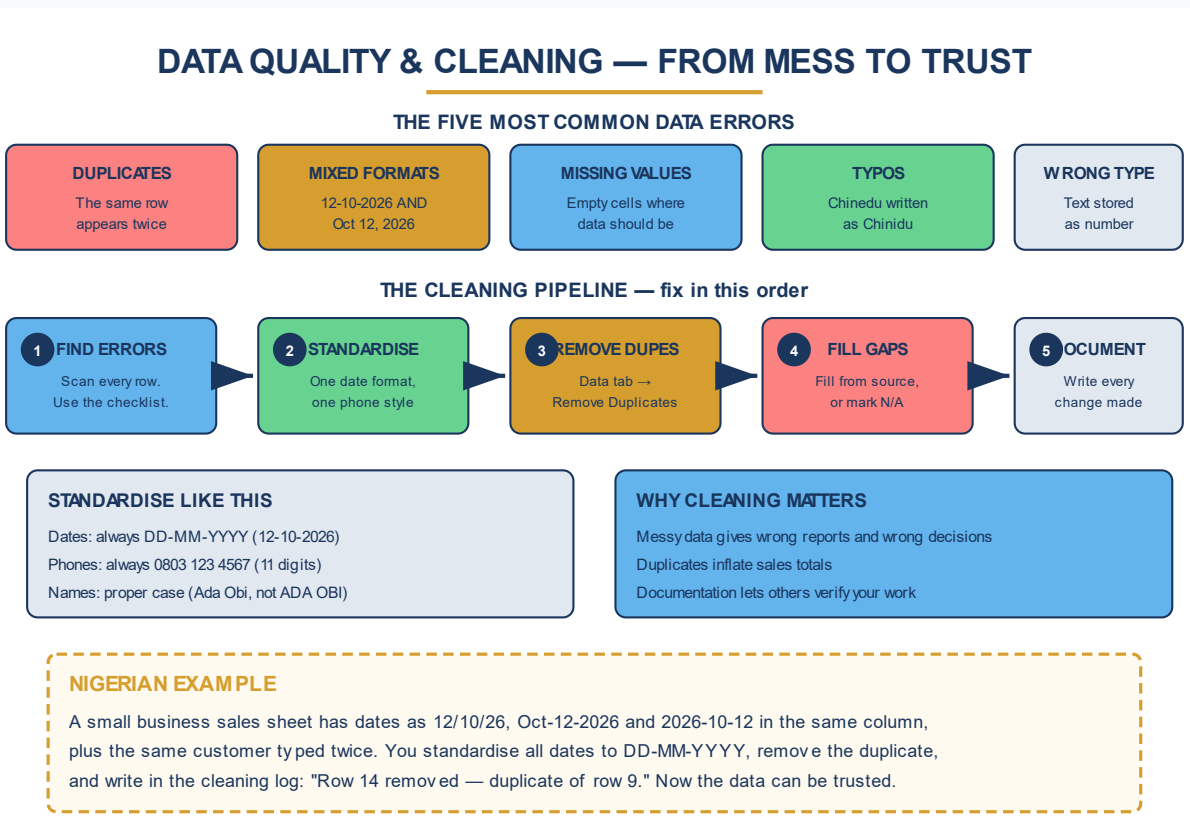
End of Session 3 — End of Week 1 (Data Entry Fundamentals)

Next Session: Session 4 — Data Validation

SESSION 04 OF 6

Data Validation

CONCEPT DIAGRAM — SESSION 04



SESSION 4: DATA VALIDATION

Week: 2 **Module:** Validation and Reconciliation **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Sessions 1–3 — Data Entry Fundamentals **Connection to Previous Session:** Builds on Session 3 by teaching data validation **Preparation Required Before Class:**

- Prepare a dataset with intentional errors for validation practice
- Prepare source documents (invoices, receipts) for cross-checking

A. SESSION IDENTIFICATION

Field	Detail
Session Number	4
Title	Data Validation
Week	2

Field	Detail
Module	Validation and Reconciliation
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Sessions 1–3
Connection to Previous Session	Builds on Session 3
Preparation	Dataset and source documents prepared

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain what data validation is** and why it matters.
2. **Check data against source documents** for accuracy.
3. **Identify discrepancies** between the dataset and source documents.
4. **Use Excel's built-in validation tools** to prevent future errors.
5. **Create a validation report** documenting all findings.
6. **Explain the difference** between validation and verification.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- Projector for instructor demonstration

Software:

- Microsoft Excel 2019 or Microsoft 365

Printed Materials:

- "Validation Checklist" (one per learner)

Sample Files:

- A dataset with intentional errors
- Source documents (invoices, receipts) for cross-checking

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"The Audit"

Instructor Setup: Display a dataset and a source document (invoice) side by side. Show where they do not match.

Instructor Script: "Good morning, everyone. Look at these two documents. On the left is a sales record in a spreadsheet. On the right is the original invoice. They should match — but they do not. The quantity is different. The price is different. The total is wrong.

This is what data validation is — checking your data against the source to make sure it is correct. Today, you will learn how to validate data like a professional auditor."

E. PREREQUISITE REVIEW (5 minutes)

Review Questions and Expected Answers:

1. "What are the five most common data errors?"
    - Inconsistent formats, misspellings, duplicates, missing values and wrong data types.
2. "How do you remove duplicates in Excel?"
    - Data tab > Remove Duplicates.
3. "What should you do if you cannot find a missing value?"
    - Mark it as 'N/A' and flag it for follow-up.
4. "Why is documentation important in data cleaning?"
    - It creates a record that others can follow and verify.
5. "What is the most important professional standard in data entry?"
    - Accuracy.

F. FULL LECTURE NOTES (35 minutes)

Topic 1: What Is Data Validation? (5 minutes)

Instructor Script: "Data validation is the process of checking data against a source to ensure it is accurate and complete. It is like auditing — you compare what you have against what should be there.

Validation vs Verification:

- **Validation:** Checking that data meets the required standards (format, type, range).
- **Verification:** Checking that data matches the original source.

Why Validation Matters:

- Catches errors before they cause problems
- Ensures data integrity
- Builds trust in the dataset
- Prevents costly mistakes"

Topic 2: Checking Data Against Source Documents (15 minutes)

Instructor Script: "Here is a systematic process for validating data against source documents:

Step 1: Prepare

- Have the dataset open in Excel.
- Have the source documents ready (invoices, receipts, forms).

Step 2: Check Each Record

- For each row in the dataset, find the matching source document.
- Compare every field: date, name, product, quantity, price, total.
- Mark any differences.

Step 3: Classify Discrepancies

- **Typo:** A digit or letter is wrong (e.g., '15000' instead of '150000').
- **Missing data:** A field is empty in the dataset but filled in the source.
- **Extra data:** A record exists in the dataset but not in the source.
- **Missing record:** A record exists in the source but not in the dataset.
- **Calculation error:** The total does not match quantity x price.

Step 4: Correct Errors

- Fix typos based on the source document.
- Fill in missing data from the source.
- Remove extra records.
- Add missing records.
- Recalculate totals.

Step 5: Document Everything

- Create a validation report listing every discrepancy found and how it was resolved."

Topic 3: Using Excel's Built-in Validation Tools (10 minutes)

Instructor Script: "Excel has tools that help prevent data entry errors.

Data Validation Rules:

1. Select the cells you want to validate.
2. Go to the Data tab.
3. Click 'Data Validation'.
4. Choose a rule type:
    - **Whole number:** Only integers allowed.
    - **Decimal:** Decimal numbers allowed.
    - **List:** Only values from a predefined list.
    - **Date:** Only dates within a range.
    - **Text length:** Limit the number of characters.
5. Set the criteria.
6. Add an error message that appears when invalid data is entered.

Conditional Formatting for Error Highlighting:

1. Select the data range.
2. Go to the Home tab.
3. Click 'Conditional Formatting'.
4. Choose a rule (e.g., 'Highlight Cells Rules' > 'Duplicate Values').
5. Duplicates are highlighted."

Topic 4: Creating a Validation Report (5 minutes)

Instructor Script: "A validation report documents everything you found and fixed during the validation process.

Validation Report Structure:

1. **Header:** Dataset name, date, validator name.
2. **Summary:** Total records checked, total errors found, total errors corrected.
3. **Error Log:** A table listing each error:
    - Record number
    - Field name
    - Error type
    - Dataset value
    - Source value
    - Resolution
4. **Sign-off:** Validator name and date."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration 1: Validating Data Against Source Documents (15 minutes)

What the Instructor Prepares:

- A dataset with intentional errors
- Source documents (invoices)

What the Instructor Says and Does: "I am going to validate this dataset against the source invoices."

1. "I start with the first record. I find the matching invoice."
2. "I compare the date: Dataset says '01/10/2026', invoice says '01/10/2026'. Match."
3. "I compare the customer name: Dataset says 'Adaeze Okafor', invoice says 'Adaeze Okafor'. Match."
4. "I compare the quantity: Dataset says '25', invoice says '2'. MISMATCH — typo in the dataset."
5. "I correct the quantity to '2'."
6. "I compare the price: Dataset says '35000', invoice says '35000'. Match."
7. "I compare the total: Dataset says '875000', but 2 x 35000 = 70000. MISMATCH — calculation error."
8. "I correct the total to '70000'."
9. "I continue this process for all 20 records."
10. "I create a validation report listing all discrepancies and resolutions."

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "Validate the Dataset"

Practical Objective: The learner will validate a dataset against source documents and create a validation report.

Instructions:

1. "Open the 'Sales Dataset' file and the 'Source Invoices' file."
2. "For each of the 20 records in the dataset:
    - Find the matching invoice.
    - Compare every field (date, name, product, quantity, price, total).
    - Mark any differences."
3. "Correct all errors in the dataset."
4. "Create a 'Validation Report' sheet that lists:
    - Record number
    - Field name
    - Error type
    - Dataset value
    - Source value
    - Resolution"
5. "Save the file as 'Validated Dataset' on the Desktop."

Completion Checklist:

- [ ] All 20 records checked against source
- [ ] All discrepancies identified
- [ ] All errors corrected
- [ ] Validation report created
- [ ] File saved as "Validated Dataset"

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "Validation Scenarios"

Answer the following questions:

1. You are validating a dataset and find a record that exists in the dataset but not in the source. What type of error is this? What do you do?
2. You find a price that is 10x higher in the dataset than in the source. What type of error is this? What do you do?
3. A field is empty in the dataset but filled in the source. What do you do?
4. You cannot find the source document for one record. What do you do?
5. What is the difference between validation and verification?

Time Allowance: 10 minutes

Complete Answer Key:

1. Extra data — remove the record from the dataset and note it in the validation report.
2. Typo (likely an extra zero) — correct the price based on the source document.
3. Fill in the missing data from the source document.
4. Mark the record as 'Unverifiable' and flag it for follow-up. Do not delete the record.
5. Validation checks that data meets required standards. Verification checks that data matches the original source.

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "The Invoice That Did Not Match"

Situation: An accountant is reconciling the sales dataset at the end of the month. She finds that the total sales in the dataset do not match the total on the bank statement. The difference is ₦150,000. She investigates and finds that one invoice was entered twice — a duplicate that was not caught during data entry.

The Problem: The duplicate was not identified during data entry or cleaning. It was only caught during reconciliation.

Why It Matters: Validation catches errors before they cause problems. Without validation, duplicates, typos and missing records can go unnoticed and cause serious consequences.

Worked Solution: The accountant removes the duplicate, corrects the total and implements a validation process to check every dataset before it is used.

Transfer Exercise: "Why is it important to validate data before using it for reports or decisions?"

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Not Checking Every Record

Symptoms: Only some records are validated. **Fix:** Always check every record against the source document.

Mistake 2: Not Documenting Discrepancies

Symptoms: Errors are corrected but not recorded. **Fix:** Always create a validation report.

Mistake 3: Correcting Based on Memory

Symptoms: The learner 'remembers' what the value should be instead of checking the source. **Fix:** Always check the source document. Never rely on memory.

Mistake 4: Ignoring Small Differences

Symptoms: The learner ignores a ₦100 difference because it is 'small'. **Fix:** Every discrepancy matters. Document and resolve all differences, no matter how small.

Mistake 5: Not Implementing Validation Rules

Symptoms: The same errors keep appearing. **Fix:** Use Excel's Data Validation tool to prevent future errors.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Validation Report"

Validate a dataset against source documents and produce a professional validation report.

Requirements:

- All records checked against source
- All discrepancies identified and classified
- All errors corrected
- Validation report created with full documentation
- File saved as "My Validation Report"

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Completeness	All records checked	Most checked	Few checked
Discrepancy identification	All found	Most found	Few found
Error correction	All corrected correctly	Most corrected	Few corrected
Report quality	Professional, complete	Mostly complete	Incomplete
File saved correctly	Saved with correct name	Saved but wrong name	Not saved

Total: 25 marks

M. KNOWLEDGE CHECK (5 minutes)

1. What is data validation? Answer: The process of checking data against a source to ensure it is accurate and complete.

2. What is the difference between validation and verification? Answer: Validation checks that data meets required standards. Verification checks that data matches the original source.

3. What are the five types of discrepancies you might find during validation? Answer: Typo, missing data, extra data, missing record and calculation error.

4. What Excel tool prevents invalid data entry? Answer: Data Validation (Data tab > Data Validation).

5. Why should you never correct data based on memory? Answer: Memory is unreliable. Always check the source document to confirm the correct value.

N. REFLECTION AND RECAP (5 minutes)

Today you learned data validation — how to check data against source documents, identify discrepancies, correct errors and create a validation report. In the next session, you will learn reconciliation and documentation.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner checks every record against the source.
- The learner identifies and classifies discrepancies correctly.
- The learner corrects errors accurately.
- The learner creates a complete validation report.

Common Misconceptions

- **"Validation is only for accountants."** Teach: anyone who works with data should validate it.
- **"Small errors do not matter."** Teach: every error matters — small errors add up.
- **"Memory is enough."** Teach: always check the source document.

Differentiation Guidance

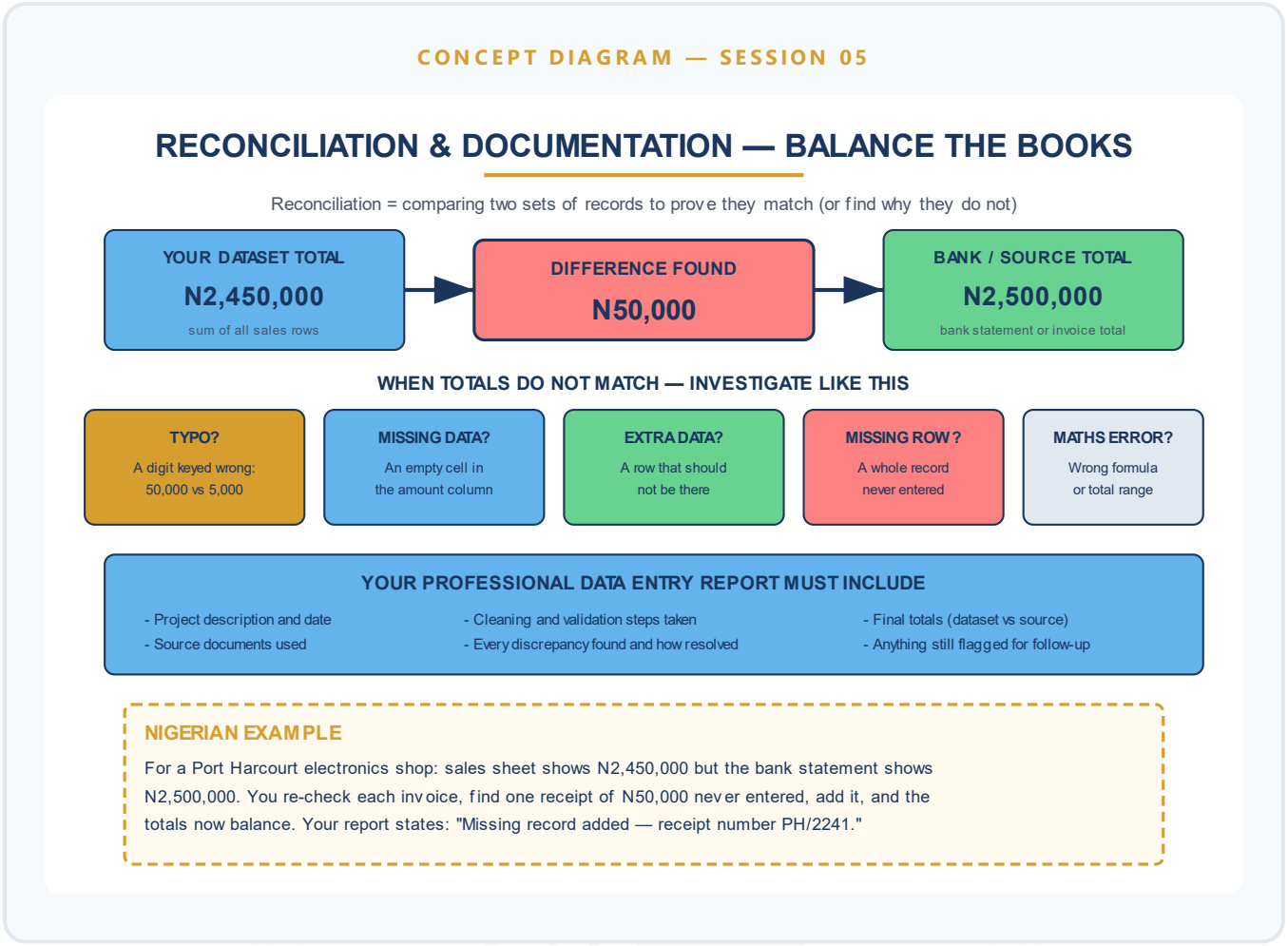
- Level 1: Provide a partially completed validation report for them to finish.
- Level 2: Complete the exercise as written.
- Level 3: Create a more complex validation scenario with multiple source documents.

End of Session 4

Next Session: Session 5 — Reconciliation and Documentation

SESSION 05 OF 6

Reconciliation Documentation



SESSION 5: RECONCILIATION AND DOCUMENTATION

Week: 2 **Module:** Validation and Reconciliation **Duration:** 105 minutes **Difficulty:** Beginner **Prerequisite Knowledge:** Sessions 1–4 — Data Entry Fundamentals, Validation **Connection to Previous Session:** Builds on Session 4 by teaching reconciliation and documentation **Preparation Required Before Class:**

- Prepare reconciliation examples with intentional discrepancies
- Prepare report templates

A. SESSION IDENTIFICATION

Field	Detail
Session Number	5
Title	Reconciliation and Documentation
Week	2

Field	Detail
Module	Validation and Reconciliation
Duration	105 minutes
Difficulty	Beginner
Prerequisite Knowledge	Sessions 1–4
Connection to Previous Session	Builds on Session 4
Preparation	Reconciliation examples, report templates

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Explain what reconciliation is** and why it matters.
2. **Compare dataset totals** against source document totals.
3. **Identify and resolve discrepancies** during reconciliation.
4. **Create a professional data entry report** documenting all processes.
5. **Apply professional documentation standards** to data entry work.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- Projector for instructor demonstration

Software:

- Microsoft Excel 2019 or Microsoft 365

Printed Materials:

- "Data Entry Report Template" (one per learner)

Internet: Not required for this session.

D. OPENING ACTIVITY (10 minutes)

"The Missing ₦50,000"

Instructor Setup: Display a dataset total and a source document total that do not match.

Instructor Script: "Good morning, everyone. The sales dataset shows total sales of ₦2,450,000. But the bank statement shows ₦2,500,000. There is a ₦50,000 difference. Where did it go? [Pause.]

Today, you will learn reconciliation — the process of comparing totals and finding out why they do not match. By the end of this session, you will be able to reconcile any dataset and produce a professional report."

E. PREREQUISITE REVIEW (5 minutes)

1. "What is data validation?" — Checking data against a source for accuracy.
2. "What are the five types of discrepancies?" — Typo, missing data, extra data, missing record, calculation error.

3. "What should you do if you cannot find a missing value?" — Mark as N/A, flag for follow-up.
4. "Why is documentation important?" — Creates a record others can follow.
5. "What Excel tool prevents invalid data entry?" — Data Validation.

F. FULL LECTURE NOTES (35 minutes)

Topic 1: What Is Reconciliation? (5 minutes)

Instructor Script: "Reconciliation is the process of comparing two sets of records to ensure they match. In data entry, this means comparing your dataset totals against source document totals — bank statements, invoices, receipts.

Why Reconciliation Matters:

- Catches errors that validation missed
- Ensures financial accuracy
- Builds trust in the dataset
- Required by auditors and accountants"

Topic 2: The Reconciliation Process (15 minutes)

Instructor Script: "Here is the systematic process for reconciling data:

Step 1: Calculate Dataset Totals

- Sum all monetary columns in the dataset.
- Count the total number of records.

Step 2: Calculate Source Document Totals

- Sum all amounts on the source documents (bank statement, invoices).
- Count the total number of source documents.

Step 3: Compare

- Subtract the dataset total from the source total.
- If the difference is zero — reconciled.
- If the difference is not zero — investigate.

Step 4: Investigate Discrepancies

- Check for missing records (in source but not dataset).
- Check for extra records (in dataset but not source).
- Check for amount differences (same record, different amount).
- Check for duplicates.

Step 5: Resolve and Document

- Correct all discrepancies.
- Document every finding and resolution."

Topic 3: Creating a Professional Data Entry Report (15 minutes)

Instructor Script: "A data entry report documents everything you did to the dataset — entry, cleaning, validation and reconciliation.

Report Structure:

1. **Header:** Dataset name, date, preparer name, department.
2. **Executive Summary:** Brief overview of what was done and the final result.
3. **Data Entry Summary:** How many records entered, from what source, using what tools.
4. **Cleaning Summary:** What errors were found and fixed during cleaning.

5. **Validation Summary:** What discrepancies were found during validation.
6. **Reconciliation Summary:** Dataset total vs source total, differences found, how resolved.
7. **Final Result:** Confirmation that the dataset is clean, validated and reconciled.
8. **Sign-off:** Preparer name, date, reviewer name, date."

G. INSTRUCTOR DEMONSTRATION (15 minutes)

Demonstration: Reconciling a Dataset (15 minutes)

What the Instructor Prepares:

- A dataset and source documents with a ₦50,000 difference

What the Instructor Says and Does:

1. "I calculate the dataset total: =SUM(F2:F21) = ₦2,450,000."
2. "I calculate the source total from the bank statement: ₦2,500,000."
3. "The difference is ₦50,000. I need to find it."
4. "I check for missing records — all 20 records are in both."
5. "I check for duplicates — none found."
6. "I compare each record individually. Record #15: dataset shows ₦100,000 but the invoice shows ₦150,000. Found it!"
7. "I correct the amount. The new total is ₦2,500,000."
8. "The dataset now matches the source. Reconciled."
9. "I create a reconciliation report documenting the process."

H. GUIDED PRACTICAL EXERCISE (20 minutes)

Exercise: "Reconcile the Dataset"

Practical Objective: The learner will reconcile a dataset against source documents and produce a reconciliation report.

Instructions:

1. "Open the 'Sales Dataset' and 'Source Documents' files."
2. "Calculate the dataset total."
3. "Calculate the source document total."
4. "Compare the totals."
5. "If there is a difference, investigate:
    - Check for missing records
    - Check for extra records
    - Check for amount differences
    - Check for duplicates"
6. "Resolve all discrepancies."
7. "Create a reconciliation report."
8. "Save as 'Reconciled Dataset'."

Completion Checklist:

- ☐ Dataset total calculated
- ☐ Source total calculated
- ☐ Totals compared
- ☐ Discrepancies investigated
- ☐ All errors corrected

- [ ] Reconciliation report created
- [ ] File saved

I. INDEPENDENT CLASS EXERCISE (10 minutes)

Exercise: "Reconciliation Scenarios"

Answer the following:

1. Dataset total: ₦1,000,000. Source total: ₦1,000,000. What is the result?
2. Dataset total: ₦500,000. Source total: ₦550,000. What do you do?
3. You find a record in the source that is not in the dataset. What type of error is this?
4. What is the final step in the reconciliation process?
5. What should a reconciliation report include?

Time Allowance: 10 minutes

Complete Answer Key:

1. Reconciled — no further action needed.
2. Investigate the ₦50,000 difference — check for missing records, amount differences or duplicates.
3. Missing record — add it to the dataset.
4. Document the reconciliation process and create a report.
5. Executive summary, data entry summary, cleaning summary, validation summary, reconciliation summary, final result and sign-off.

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "The Audit"

Situation: An external auditor arrives at a company to review their financial records. The auditor asks for the sales dataset and the data entry documentation. The administrator provides a clean dataset with a full report — entry log, cleaning log, validation report and reconciliation report. The auditor is impressed and completes the audit quickly with no issues.

The Problem: Without documentation, the auditor would have to re-check every record, taking weeks instead of days.

Relevant Concept: Documentation. Professional documentation saves time, builds trust and satisfies auditors.

Worked Solution: The administrator creates a complete data entry report for every dataset. The audit is smooth and the company passes with no issues.

Transfer Exercise: "Why do auditors need documentation of data entry processes?"

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Not Calculating Totals Accurately

Fix: Use Excel formulas (=SUM) rather than manual addition.

Mistake 2: Stopping at the First Discrepancy

Fix: Continue investigating until ALL discrepancies are resolved.

Mistake 3: Not Documenting the Reconciliation Process

Fix: Always create a reconciliation report.

Mistake 4: Reconciling Without Source Documents

Fix: Never reconcile without the original source documents.

Mistake 5: Not Getting a Second Person to Review

Fix: Have someone else review your reconciliation before finalising.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Data Entry Report"

Create a professional data entry report for the work done in Sessions 1–5.

Requirements:

- Executive summary
- Data entry summary
- Cleaning summary
- Validation summary
- Reconciliation summary
- Final result confirmation
- Sign-off section

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Completeness	All sections present	Most present	Several missing
Accuracy	All information correct	Mostly correct	Several errors
Professional quality	Professional, well-written	Mostly professional	Informal
Sign-off	Complete and correct	Present but incomplete	Missing
File saved correctly	Saved with correct name	Saved but wrong name	Not saved

Total: 25 marks

M. KNOWLEDGE CHECK (5 minutes)

1. What is reconciliation? Answer: The process of comparing two sets of records to ensure they match.

2. What is the first step in the reconciliation process? Answer: Calculate the dataset totals.

3. What should you do if the dataset total does not match the source total? Answer: Investigate — check for missing records, extra records, amount differences and duplicates.

4. What should a data entry report include? Answer: Executive summary, data entry summary, cleaning summary, validation summary, reconciliation summary, final result and sign-off.

5. Why is documentation important? Answer: It creates a record that others can follow, verify and audit.

N. REFLECTION AND RECAP (5 minutes)

Today you learned reconciliation and documentation — how to compare totals, investigate discrepancies and produce a professional data entry report. In the next session — the final session — you will complete your capstone project.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner calculates totals accurately.
- The learner investigates discrepancies systematically.
- The learner resolves all errors.
- The learner creates a professional report.

Common Misconceptions

- **"Reconciliation is only for accountants."** Teach: anyone who works with financial data should reconcile.
- **"Documentation is bureaucratic nonsense."** Teach: documentation protects you and builds trust.
- **"If the totals are close enough, it is fine."** Teach: the totals must match exactly.

Differentiation Guidance

- Level 1: Provide a partially completed reconciliation report.
- Level 2: Complete the exercise as written.
- Level 3: Reconcile a more complex dataset with multiple source documents.

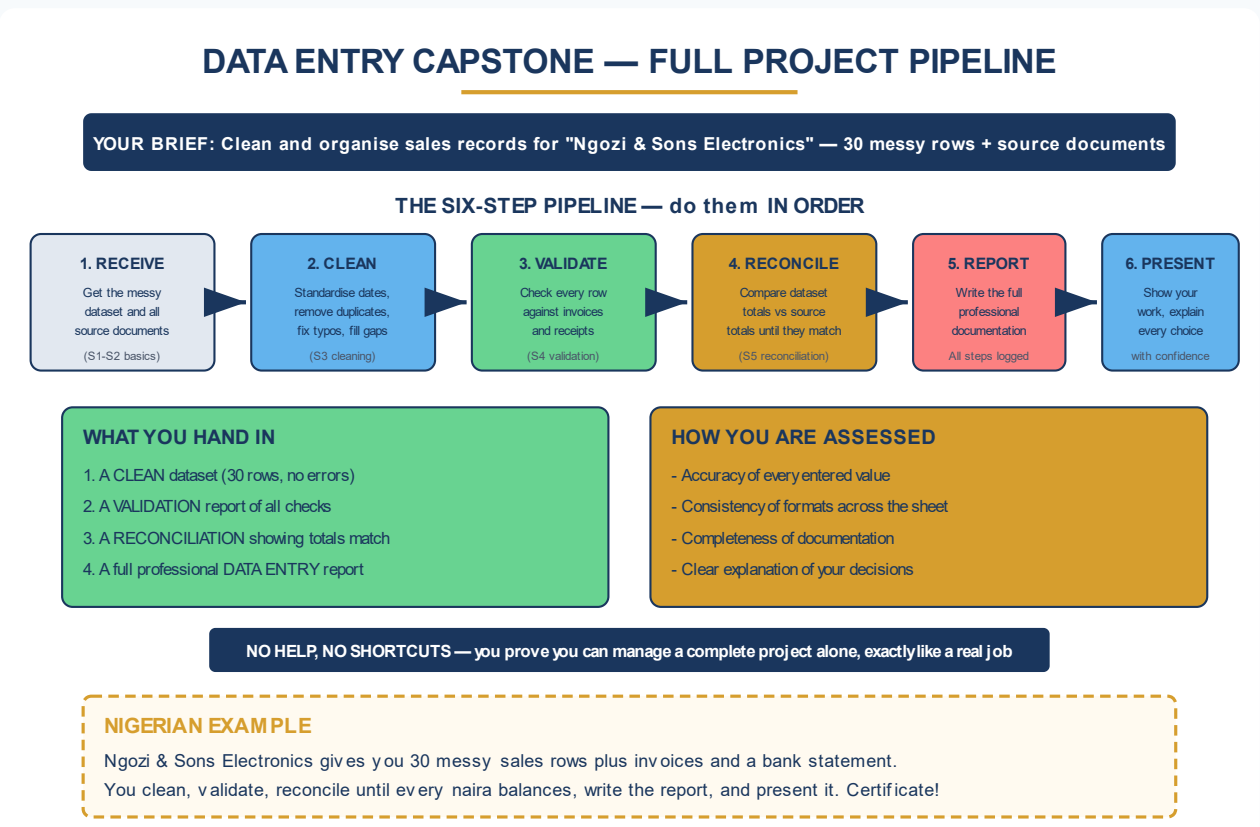
End of Session 5

Next Session: Session 6 — Final Project (Capstone)

SESSION 06 OF 6

Final Project Capstone

CONCEPT DIAGRAM — SESSION 06



SESSION 6: FINAL PROJECT — DATA ENTRY CAPSTONE

Week: 2 **Module:** Validation and Reconciliation **Duration:** 105 minutes **Difficulty:** Beginner (tests all accumulated skills)
Prerequisite Knowledge: Sessions 1–5 — All prior knowledge **Connection to Previous Session:** Integrates and tests everything taught in Sessions 1–5 **Preparation Required Before Class:**

- Prepare capstone datasets (raw data + source documents)
- Print assessment rubrics and certificate sign-off sheets

A. SESSION IDENTIFICATION

Field	Detail
Session Number	6
Title	Final Project — Data Entry Capstone
Week	2

Field	Detail
Module	Validation and Reconciliation
Duration	105 minutes
Difficulty	Beginner (tests all accumulated skills)
Prerequisite Knowledge	Sessions 1–5
Connection to Previous Session	Integrates and tests everything
Preparation	Capstone datasets, rubrics, certificates

B. LEARNING OUTCOMES

By the end of this session, the learner will be able to:

1. **Clean a messy dataset** — standardise formats, remove duplicates, correct errors.
2. **Validate against source documents** — identify and resolve discrepancies.
3. **Reconcile totals** — compare dataset totals with source document totals.
4. **Produce a professional data entry report** documenting all processes.
5. **Present the work** to the instructor with confidence.
6. **Explain all decisions** made during the project.

C. REQUIRED MATERIALS

Hardware:

- One desktop computer per learner
- Projector for presentation

Software:

- Microsoft Excel 2019 or Microsoft 365

Printed Materials:

- Capstone brief (one per learner)
- Assessment rubric (one per learner)
- Certificate sign-off sheet (one per learner)

Sample Files:

- Raw messy dataset (30 rows with intentional errors)
- Source documents (invoices, receipts, bank statement)

Internet: Not required for this session.

D. OPENING ACTIVITY (5 minutes)

Instructor Script: "Good morning, everyone. This is it — the final session. Over the past five sessions, you have learned data entry fundamentals, spreadsheet data entry, data cleaning, validation and reconciliation. Today, you will prove that you can manage a complete data entry project from start to finish.

I am not going to help you. I am going to give you a messy dataset and source documents, and you will produce a clean, validated, reconciled dataset with a full professional report. This is your moment to show what you have learned."

E. PREREQUISITE REVIEW (5 minutes)

1. "What are the four main data types?" — Text, numbers, dates and currency.
2. "What is the most efficient way to enter data?" — Column by column, using keyboard shortcuts.
3. "What are the five most common data errors?" — Inconsistent formats, misspellings, duplicates, missing values and wrong data types.
4. "What is the difference between validation and verification?" — Validation checks standards. Verification checks against source.
5. "What is reconciliation?" — Comparing two sets of records to ensure they match.

F. FULL LECTURE NOTES (10 minutes)

The Capstone Project Explained

Instructor Script: "Here is what you are going to do:

The Project: You are a data entry professional hired by 'Ngozi & Sons Electronics' to clean and organise their sales records. You will receive:

- A messy dataset (30 rows with errors)
- Source documents (invoices and a bank statement)

Your Tasks:

1. Clean the dataset — standardise formats, remove duplicates, correct errors, fill missing values.
2. Validate against source documents — check every record against the invoices.
3. Reconcile totals — compare the dataset total with the bank statement total.
4. Produce a professional data entry report documenting all processes.
5. Present your work to the instructor.

Deliverables:

- Cleaned and validated Excel workbook
- Data entry report (Word document)
- Presentation to the instructor"

G. INSTRUCTOR DEMONSTRATION (0 minutes)

There is no instructor demonstration in this session.

H. GUIDED PRACTICAL EXERCISE (0 minutes)

There is no guided practical in this session.

I. INDEPENDENT CLASS EXERCISE (70 minutes)

Exercise: "Data Entry Capstone"

Scenario: "You are a data entry professional. You have received a messy dataset and source documents. Complete the project from raw data to final deliverable."

Task: Complete the following project without assistance:

Step 1: Clean the Dataset (20 minutes)

1. Open the 'Raw Sales Data' file.
2. Identify all errors.
3. Standardise all formats (dates, names, currency).
4. Remove duplicates.
5. Fill missing values or mark as N/A.
6. Correct all typos and calculation errors.
7. Create a cleaning log.

Step 2: Validate Against Source Documents (20 minutes)

1. Open the 'Source Invoices' file.
2. Check every record in the dataset against the matching invoice.
3. Identify and correct all discrepancies.
4. Create a validation report.

Step 3: Reconcile Totals (15 minutes)

1. Calculate the dataset total.
2. Calculate the bank statement total.
3. Compare the totals.
4. Investigate and resolve any differences.
5. Create a reconciliation report.

Step 4: Produce the Data Entry Report (10 minutes) Create a Word document that includes:

- Executive summary
- Data entry summary
- Cleaning summary
- Validation summary
- Reconciliation summary
- Final result
- Sign-off

Step 5: Present (5 minutes) Present your work to the instructor. Explain:

- What errors you found
- How you fixed them
- What the final result is
- What you learned from the process

Requirements:

- Complete all five steps independently.
- All errors must be corrected.
- All totals must reconcile.
- Report must be professional and complete.

Time Allowance: 70 minutes

Success Criteria:

- Dataset is clean and validated.
- Totals reconcile.
- Report is complete and professional.
- Presentation is clear and confident.

J. REAL-WORLD CASE STUDY (5 minutes)

Case: "Emmanuel's First Client"

Situation: Emmanuel completed the Data Entry course at CEA. He passed the capstone assessment. Two weeks later, a friend asks him to clean and organise a messy sales dataset for their small business. Emmanuel sits down and completes the entire project — cleaning, validation, reconciliation and documentation — in one afternoon. His friend is thrilled.

The Problem: Emmanuel's friend had a messy dataset and did not know how to clean it.

Relevant Concept: All the skills taught in Sessions 1–5.

Worked Solution: Emmanuel succeeded because he had practised every step of the process multiple times.

Transfer Exercise: "Think about a business that has messy data. Write down the steps you would take to clean and organise it."

K. COMMON MISTAKES AND TROUBLESHOOTING (5 minutes)

Mistake 1: Not Checking Every Record

Fix: Always check every record against the source.

Mistake 2: Not Documenting the Process

Fix: Always create a report.

Mistake 3: Not Reconciling Totals

Fix: Always compare dataset totals with source totals.

Mistake 4: Presenting Without Preparation

Fix: Review your work before presenting. Be ready to explain your decisions.

Mistake 5: Leaving the Dataset Messy

Fix: Always leave the dataset clean, organised and professional.

L. SESSION ASSIGNMENT (5 minutes)

Assignment Title: "My Learning Reflection"

Write a reflection on the entire Data Entry course.

Requirements:

- Three things you learned
- Two things you found challenging
- One thing you want to learn next
- How you feel now compared to Session 1

Marking Rubric:

Criterion	Excellent (5)	Good (3)	Needs Improvement (1)
Three things learned	Specific, accurate	Mostly specific	Vague
Two challenges	Specific, honest	Mostly specific	Vague
One next step	Clear, specific	Present but vague	Missing
Feeling comparison	Insightful, honest	Present but generic	Missing
File saved correctly	Saved on Desktop	Saved but wrong location	Not saved

Total: 25 marks

M. KNOWLEDGE CHECK (15 minutes)

Quiz

Section A: Fundamentals (5 questions)

1. What are the four main data types?
2. What is the most efficient way to enter data into Excel?
3. What does AutoFill do?
4. What is the most important professional standard in data entry?
5. What are the five most common data errors?

Section B: Cleaning (5 questions) 6. How do you remove duplicates in Excel? 7. What should you do if you cannot find a missing value? 8. What Excel function converts text to Title Case? 9. Why is it important to document data cleaning changes? 10. What is the standard date format in Nigeria?

Section C: Validation and Reconciliation (5 questions) 11. What is data validation? 12. What is the difference between validation and verification? 13. What are the five types of discrepancies? 14. What is reconciliation? 15. What should a data entry report include?

Section D: Integration (5 questions) 16. What are the six steps of the data entry process? 17. Why should you never make up missing data? 18. What should you do if dataset totals do not match source totals? 19. Why is documentation important for auditors? 20. What is the most important skill in data entry?

Answer Key

Section A:

1. Text, numbers, dates and currency.
2. Column by column, using keyboard shortcuts.
3. Automatically fills in repetitive data based on a pattern.
4. Accuracy.
5. Inconsistent formats, misspellings, duplicates, missing values and wrong data types.

Section B: 6. Data tab > Remove Duplicates. 7. Mark as N/A and flag for follow-up. 8. PROPER 9. It creates a record others can follow and verify. 10. DD/MM/YYYY

Section C: 11. The process of checking data against a source to ensure it is accurate and complete. 12. Validation checks that data meets required standards. Verification checks that data matches the original source. 13. Typo, missing data, extra data, missing record and calculation error. 14. The process of comparing two sets of records to ensure they match. 15. Executive summary, data entry summary, cleaning summary, validation summary, reconciliation summary, final result and sign-off.

Section D: 16. Plan, enter, clean, validate, reconcile, document. 17. Making up data is dishonest and can lead to wrong decisions. 18. Investigate — check for missing records, extra records, amount differences and duplicates. 19. It creates a record that auditors can follow and verify. 20. Accuracy.

N. REFLECTION AND RECAP (5 minutes)

Plain-English Summary

Today you completed a full data entry project — cleaning, validating, reconciling and documenting a messy business dataset. You have completed the Data Entry course. You should be proud of yourself.

Five Important Takeaways from the Entire Course

1. Accuracy is the foundation of all data entry work.

2. Efficiency tools — shortcuts, AutoFill, column-by-column entry — make you more productive.
3. Data cleaning is essential — never use data without cleaning it first.
4. Validation and reconciliation catch errors before they cause problems.
5. Professional documentation builds trust and satisfies auditors.

Questions That Help You Recognise What You Can Now Do

- Can you enter data efficiently into Excel?
- Can you clean a messy dataset?
- Can you validate data against source documents?
- Can you reconcile totals?
- Can you produce a professional data entry report?
- Can you explain your decisions?

What Comes Next

You have completed the Data Entry course. You are now ready for the next level. At Cyber Elias Academy, you can continue with:

- **Microsoft Office** — Learn Word, Excel and PowerPoint in depth.
- **Digital Productivity** — Learn Google Workspace and cloud-based data management.
- **Computer Repairs** — Learn to diagnose and fix computer problems.
- **Data Analysis** — Learn to analyse and visualise data.

O. INSTRUCTOR EVALUATION NOTES

Observable Signs of Competence

- The learner cleans the dataset thoroughly.
- The learner validates every record against the source.
- The learner reconciles totals accurately.
- The learner produces a professional report.
- The learner presents with confidence.

Common Misconceptions

- **"I can just learn this on my own."** Teach: structured learning saves time and prevents bad habits.
- **"Data entry is a dead-end job."** Teach: data entry is the foundation of data management, analysis and business intelligence.
- **"I am too old to learn this."** Teach: age is not a barrier to learning data entry skills.

Suggested Remedial Exercises

- For learners who do not pass the capstone: Allow them to practise on academy machines during opening hours. Let them retake the assessment when ready.
- For learners who cannot clean thoroughly: Provide additional cleaning exercises with guided steps.
- For learners who cannot reconcile: Walk them through the reconciliation process step by step.

Differentiation Guidance

Level 1 — Supported:

- Allow extra time for the capstone.
- Provide partially completed templates.
- Offer one-on-one support.
- Allow retake after additional practice.

Level 2 — Independent:

- Complete the capstone as written.
- Answer all knowledge check questions.
- Complete the reflection assignment.

Level 3 — Challenge:

- Add additional error types to the dataset.
- Create a more complex report with charts and analysis.
- Reconcile multiple datasets simultaneously.

CERTIFICATE SIGN-OFF

Learner Name: _____ **Date:** _____ **Capstone Result:** Pass / Not Yet Competent **Knowledge Check Score:** _____ / 100 **Typing Speed (Session 1):** _____ WPM **Typing Speed (Session 6):** _____ WPM **Improvement:** _____ WPM

Deliverables Completed:

- ☐ Cleaned and validated Excel workbook
- ☐ Data entry report (Word document)
- ☐ Presentation to the instructor

Instructor Signature: _____ **Certificate Awarded:** Yes / No

Comments:

End of Session 6 — End of Course

Congratulations! You have completed the Data Entry course at Cyber Elias Academy.

Next Step: Continue your learning with Microsoft Office, Digital Productivity, or another CEA course.